ADNOC GAS OPERATIONS & MARKETING LLC

ADDENDUM TO DGS-6300-004 PIPELINE WELDING SPECIFICATIONS

EPCm CONTRACT NO.: 3000076501
ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE

PROJECT 5.0 (P5)

EPCm FOR HABSHAN DBN

REV.	DESCRIPTION	DATE	PREPARED	CHECKED	APPROVED
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1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including below new pipelines
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no.s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006) for complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE-GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design basis.

2.0 PURPOSE OF THIS DOCUMENT

The purpose of this document, together with its attachments, is to define the minimum requirements for fabrication, field welding and inspection works of gas/liquid pipelines under the PROJECT scope.

It covers the welding and NDE of pipe-to-pipe, pipe to hot bend, pipe to fitting and fitting-to-fitting welds, including multiple joints, tie-ins and attachments.

For pipelines in H_2S sour services, additional requirements from the Addendum to AGES-SP-07-003 (30201_50150_LT_HE-ADD-SP-07-003) shall be met.

This document shall be read in conjunction with the attached COMPANY Specification DGS-6300-004 Rev.3.

3.0 DEFINITIONS

3.1 GENERAL DEFINITIONS

For this document, the following general definitions are applicable.

Table 3-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by the EPCm CONTRACTOR.
EPCm CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipelines.
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the COMPANY
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE stage (EPCm) of the PROJECT.
VENDOR	The party(s) which manufactures and/or supplies piping, equipment and services to perform the duties specified
SHALL	The use of the word “shall” indicate a mandatory requirement
SHOULD	The use of the word “should” indicate a strong recommendation to comply with the requirements of this document
EPCm CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.

DEFINITION	DESCRIPTION
SUB-CONTRACTOR	The COMPANY approved party, which carries out services subcontracted by EPCm CONTRACTOR.

3.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 3-2: Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas

DEFINITION	DESCRIPTION
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields. Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Upstream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles

4.0 AMENDMENTS

The following amendments and additions shall apply to COMPANY DGS-6300-004 (Rev. 3) Design General Specification Pipeline Welding.

In addition to the DGS-6300-004 (Rev. 3), the referenced project specifications included in Section 4.2 of this document shall be followed.

Clauses not amended shall be considered applicable requirements of the specification.

Some of the amendments are specific to this project as mentioned below and shall be considered as supplement/ amendment to DGS-6300-004 (Rev. 3).

Wherever amendments have been specified, these are noted with guide words such as "REPLACE", "ADD", "DELETE" or "MODIFY.

(Add) New paragraph or supplemental requirements and/or clarifications added to an existing paragraph.

(Replace) Replace the complete paragraph / section.

(Modify) Where the words in an existing clause have been modified but not a substitution that invalidates the paragraph.

(Delete) Where the clause no longer applies and is removed, either in full or in part.

4.1 Amendments to Section 2.0 – Codes and Standards

AMERICAN WELDING SOCIETY (AWS)

- Add: AWS A4.3 – Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

- Replace: “ASTM 370” with “ASTM A370”
- ASTM G39 – Replace title with “Standard Practice for Preparation and Use of Bent-Beam Stress-Corrosion Test Specimens”
- ASTM E92 – Replace title with: “Standard Test Methods for Vickers Hardness and Knoop Hardness of Metallic Materials"

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- ASME B31.4 – Replace title with: “Pipeline Transportation Systems for Liquid and Slurries”
- Add the following:

  ASME BPVC Section V: Non-Destructive Examination

  ASTM E 1961: Standard Practice for Mechanized Ultrasonic Examination of Girth Welds Using Zonal Discrimination with Focused Search Units

AMERICAN PETROLEUM INSTITUTE (API)

- Replace API SPEC 5L – Replace title with: “Line Pipe, 46th Edition”

BRITISH STANDARDS (BS)

- Replace: “BS 7448, Part 1” with: “BS EN ISO 12737-2010 - Metallic materials - Determination of Plane-strain Fracture Toughness”.
- Delete: “BS 7448 Part 2”.

4.2 Amendments to Section 3.0 – Reference Document

DESIGN GENERAL SPECIFICATION (DGS) / ADNOC GROUP ENGINEERING SPECIFICATION (AGES)

- Remove the following:
  - DGS-000-002 Material for use in H2S Containing Environment.
  - DGS 9530-001 Factory Hot Bends
  - DGS 9530-002 Pipeline Pigging Tees
  - DGS 9540-001 Pipeline Scrapper Launcher and Receivers
  - AGES-SP-09-009 Piping Tie-ins and Hot Tapping
  - AGES-SP-10-002 Specification for Subsea Pipeline systems
  - AGES-SP-10-001 Specification for Line Pipe
  - AGES-SP-10-003 Onshore Pipelines Design & Construction Specification
- Add the following:
  - AGES-SP-10-009 for Pipeline Launcher and Receivers
  - AGES-SP-10-010 Hot Induction Bend Specification
  - AGES-SP-10-011 Pipeline Pigging Tees Specification
  - AGES-SP-10-013 Isolation Joints
  - AGES-SP-10-015 Specification for Anchor Flanges
- Add the following:

  PROJECT SPECIFIC REFERENCES

  - BOD-30201_50150_LT_HE-PL-000-95-00-001 – Pipeline Design Basis.
  - 30201_50150_LT_HE-ADD-SP-07-003 Addendum to AGES-SP-07-003 Requirement for Materials in Severe Service
  - 30201_50150_LT_HE-ADD-SP-10-006 Addendum to AGES-SP-10-006 CRA Clad and Lined Steel Pipe (Amendments Supplements to API Spec 5LD)
  - 30201_50150_LT_HE-ADD-SP-10-001 Addendum to AGES-SP-10-001 Specification for Linepipe.
  - 30201_50150_LT_HE-ADD-SP-10-003 Addendum to AGES-SP-10-003 Onshore Pipelines Design & Construction Specification.
  - 30201_50150_LT_HE-ADD-SP-10-014 Addendum to AGES-SP-10-014 Specification for Pipeline Cleaning, Gauging, Filling and Hydrostatic Testing.

- 30201_50150_LT_HE-ADD-SP-09-003 Addendum to AGES-SP-09-003 for Manual Piping and Pipeline Valves Specification
- 30201_50150_LT_HE-ADD-9510-004 Addendum to DGS 9510-004 Specification for Onsite, Non Piggable Piping Materials

4.3 Amendments to Section 7.0 – Design Considerations

- Replace whole paragraph with following:

  PROJECT specific design conditions are described in document below:

  - IE-30201_50150_LT_HE-GG-000-50-00-001 – Project Design Data.
  - BOD-30201_50150_LT_HE-GG-000-00-00-001 – Process Design Basis.
  - BOD-30201_50150_LT_HE-GG-000-00-00-002 – Utility Design Basis.
  - BOD-30201_50150_LT_HE-PL-000-95-00-001 – Pipeline Design Basis.

4.4 Amendments to Section 8.0 – Welding Equipment, Tools and Supplies

- Add the following at the end of second paragraph:

  Standard calibration shall be performed for welding equipment including its instrumentation (i.e., arc monitoring, current meters, temperature measurement devices, etc.) at once in every six months. The valid calibration log and certificate(s) shall be maintained and made available upon COMPANY request. In addition, Welder and welding operator shall ensure the routine self-calibration check of welding equipment parameter during welding operation.

  For semi-automatic and mechanized/automatic welding, the welding power supply and welding bugs used in production weld shall be of the same type, model and software version as those used during welding procedure qualification.

- Replace first sentence of third paragraph with:

  A detailed procedure for operation of Welding, Automatic (not SAW) or Manual, outlining equipment's, set-up and sequence of operation shall be submitted for COMPANY approval.

- Add the following at the end of section 8.0:

  Use of Copper backing strips (or) copper back-up rings mounted on internal clamps is not allowed in all welding process, including GMAW.

4.5 Amendments to Section 9.0 – Pipeline Materials

- Replace first sentence with following:

  Line pipe shall be in accordance with Addendum to AGES-SP-10-001 Specification for line pipe Document No. 30201_50150_LT_HE-ADD-SP-10-001 and Addendum to

AGES-SP-10-006 CRA Clad and Lined Steel Pipe (Amendments Supplements to API Spec 5LD) Document No. 30201_50150_LT_HE-ADD-SP-10-006. Reference to the Project Design Basis and any other relevant PROJECT specifications are required for further line pipe information (i.e., sour service, material grade, minimum design temperature, etc.). HFW pipes might be used for pipe sizes up to 24-inch, however, shall not be used in sour service conditions.

- Replace: “Line pipe residual magnetism may be 25 gauss maximum.”

  With: Line pipe residual magnetism may be 20 gauss maximum.

4.6 Amendments to Section 10.0 – Electrodes, Filler Wire and Fluxes

- Add following at the end of section 10:

  Welding consumables shall be selected based on their mechanical properties, compatibility with the parent materials, their suitability for the intended service, and consideration of polarity, position, and direction of welding. This shall be demonstrated by performing a full set of mechanical test including all-weld metal tensile test. Only the batches of consumables tested shall be used for production welding. Additional testing shall be conducted to attain COMPANY approval for additional batches of consumables. The additional testing shall be undertaken on the weld made in accordance with the production WPS.

  COMPANY reserves the right to request BS EN 10204 Type 3.2 certification of any welding consumables as and when deemed necessary.

  Under-matching consumable is not allowed. EPCm CONSTRUCTION CONTRACTOR to ensure all the consumable used has tensile and yield strength exceeding the minimum required for the material being welded.

  For low hydrogen SMAW consumables, MANUFACTURER’s certificate shall include average-diffusible-hydrogen level as determined and reported in accordance with AWS A4.3. The weld metal diffusible hydrogen content for welding consumables shall be 5 ml/100gr of weld maximum.

  The use of permanent or consumable backing ring is not permitted. Non-consumable temporary backing as an integral permanent feature on an internal clamp may be permitted subject to satisfactory demonstration of the process.

  Portable welding electrode quiver and oven shall have valid calibration prior to use.

4.7 Amendments to Section 12.0 – Qualification of Welding Procedures

Sub Section 12.1 General

- Replace: “Welding procedures are required for all production welding including butt welds, fillet welds, and all associated repairs”.

  With: PROJECT specific Welding Procedure Specification (WPS) in accordance with API

1104 latest edition as supplemented by this document are required for all production welding including butt welds, fillet welds and all associated repairs. Previous welding procedure qualifications are not accepted.

- Replace third paragraph, third bullet: “Welding of fittings”.

  With: Welding of fittings (i.e., Hot Induction Bends, Pup Pieces, Anchor Flanges, Isolation Joints, Pig Traps Pig Launcher and Pig Receiver, Barred Tees)

- Replace: “Gas Metal Arc Welding (GMAW) – Automatic, subject to EPCm CONSTRUCTION CONTRACTOR having a successful past history using the proposed automatic GMAW process and/or system and the following:”

  With: Gas Metal Arc Welding (GMAW) – Automatic-mechanized welding shall be carried out preferably by reputable and well recognized Automatic Welding Specialist Subcontractor having successful history of more than 5-years carrying out minimum 500 kms automatic pipeline girth welds with repair rate less than 2% and the following:

- Replace: “- 100% UT testing, in addition to RT, is carried out on the pre-weld qualification test welds.”

  With: 100% UT or Automatic Ultrasonic Testing (AUT), in addition to RT, is to be carried out on the pre-weld qualification test welds. For the pipeline with more than 25 kilometers in length, AUT shall be carried out.

- Replace: “Oxyacetylene welding and flash butt welding process shall not be permitted.”

  With: Oxyacetylene welding, flash butt welding and FCAW welding processes shall not be permitted.

- Add following at the end of sub section 12.1:

  Qualification of welding procedure for automatic welding shall be based on the actual field welding process to be adopted by the EPCm CONSTRUCTION CONTRACTOR in respect of the following:

  - Number of welders per joint;
  - Number of machines per joint;
  - Sequence of welding

  Any change to the above parameters during production welding shall require re-qualification of the procedure. If more than one machine per joint is to be used for WPS qualification it must simulate the actual site sequence.

  Direct current negative polarity root passes are not allowed when using the SMAW process with vertical down technique. All fillet welds shall be made using low hydrogen electrodes.

  For automatic UT and conventional UT inspection, a detailed procedure shall be

prepared. Prequalification on a test girth weld made with known defects of required sizes as per sensitivity requirement using the same pipe grade, OD and thicknesses of the line pipe of the main line project.

For pipes and fittings having unequal wall thickness, the welding ends shall meet the preparation as defined in ASME B31.8, Appendix I, Fig. I-5.

When the pipes of same material grades are supplied by more than one supplier, the welding procedure specification and qualification requirement shall require prior approval by COMPANY.

For fully automatic-mechanized welding system (i.e., Automatic GMAW) where the EPCm CONSTRUCTION CONTRACTOR previous experience is limited or where the system will be used under new conditions shall require a more extensive prequalification programme or documentation before they can be accepted for use. The extent and the contents of a prequalification programme for such mechanized welding systems shall be agreed upon before it is commenced. The EPCm CONSTRUCTION CONTRACTOR shall prove and document to the COMPANY that the welding systems are reliable and that the process can be continuously monitored and controlled.

EPCm CONSTRUCTION CONTRACTOR approved Welding Procedure Specification (WPS) is valid in field/site (as specified in the PROJECT) and workshops operated by EPCm CONSTRUCTION CONTRACTOR under identical or equivalent technical and quality control. Welding procedures shall not be transferred from EPCm CONSTRUCTION CONTRACTOR to another party without requalification.

The relevant COMPANY approved WPSs shall be written in English and be displayed at each work location and shall be readily available to each welder and welding operator.

Sub Section 12.2 Records

- Add following at the end of sub section 12.2:

  As minimum, the welding procedure documentation package for each welding procedure shall contain the following information/data:

  - Geometric details of joint preparation;
  - Alignment method & tacking details;
  - Record of welding parameter data (WPQR);
  - Weld preparation diagram/sketch & run sequence;
  - NDT results;
  - All mechanical test results including photo-macrographs, showing locations of hardness indents;
  - Weld metal chemical composition test report;
  - Corrosion testing report, as necessary;

- Welding consumable and gas certificate;
- Mill-test certificate of base material;
- Heat treatment charts (if applicable);
- Details of equipment used in the WPQT.

The following data shall also be available if requested by COMPANY:

- Calibration certificates for equipment and measuring devices;
- NDT procedures/written practices;
- Welding Inspector and NDT personnel certificates;
- NDT procedure qualification records;

Sub Section 12.3 Weld Procedure Specification

- Replace: "In addition to the requirements of Section 5.3 of API 1104, the following minimum details shall be included on the Welding Procedure Specification:"

  With: In addition to the requirements of Section 5.3 & 12.4 (for automatic-mechanized welding) of API 1104 latest edition, the following minimum details shall be included on the Welding Procedure Specification:

  1. Project identifications.
  2. Consumable trade names, type and lot number.
  3. Weld preparation and fit up tolerances.
  4. Preheat temperature and method.
  5. Time lapse between end of root bead and start of hot pass (second pass).
  6. Interpass temperature (minimum).
  7. Nozzle diameter for automatic welding.
  8. Oscillation width and frequency for automatic welding.
  9. Electrode stick-out length.
  10. Welding equipment type and model
  11. Wire feed speed.

Sub Section 12.4 Weld Procedure Qualification

- Replace: "Welding procedures shall be qualified under simulated conditions similar to the planned installation, using the same welding, alignment and pre-heat equipment that will be used in production. Sufficient test welds shall be made to include repair procedure qualification and all required mechanical testing."

  With: Welding procedures shall be qualified under simulated conditions similar to the

planned installation, using the same welding equipment, similar alignment and preheat equipment that will be used in the production. Sufficient test welds shall be made to include repair procedure qualification, and all required mechanical testing. Minimum Preheat of 75ºC shall be done for grade 65 KSI and above grade.

- Replace: “In addition to those in Section 5.4 of API 1104 the following essential variables shall be added:”

  With: In addition to those in Section 5.4 and 12.5 (for automatic-mechanized welding) of API 1104 latest edition, the following essential variables shall be added:

- Replace sub-paragraph 12.4.(c): “A change in grade of pipe, diameter, or wall thickness grouping (as identified in API 1104 paragraph 6.2.2 Items D and E) or source of supply or manufacturer. For automatic GMAW, each change in wall thickness.”

  With: A change in grade of pipe or wall thickness grouping (as identified in API 1104 paragraph 5.4.2.2) and change in outside diameter grouping (as identified in API 1104 paragraph 6.2.2 items D) or source of material supply or MANUFACTURER. For automatic GMAW, any change in grouping requirements as identified in API 1104 latest edition paragraph 12.5.2.2.

- Add at the end of sub-paragraph 12.4:

  s) Any interruption in the welding (incomplete weld / overnight). Or an increase in the time between the completion of the root bead and the start of the second bead, this is not required for low hydrogen welding processes.

  t) Any change in lot/batch number of the filler wire for automatic welding.

- Replace: “All three joints shall pass the required NDE. One of the welds shall be used for repair qualifications.”

  With: EPCm CONSTRUCTION CONTRACTOR to ensure the number of weld joints are sufficient for all qualification testing required, this including weld for repair welding qualification. All joints shall pass the required NDE prior destructive testing.

- Add the following at the end of sub section 12.4:

  For all automatic-mechanized welding, the welding qualification shall include all applicable equipment and controller parameter settings that will be used during production welding.

Sub Section 12.5 Weld Procedure Qualification Testing

- Replace sub paragraph a): “The welds shall be visually inspected and then fully X-radiographed prior to destructive tests.”

  With: The welds shall be visually inspected and then fully X- radiographed prior to destructive tests. In addition, Automatic GMAW PQR coupons shall also be tested to

AUT.

- Replace the last sentence in sub paragraph a): ”GMAW automatic welding shall be additionally fully inspected in accordance with an ultrasonic testing procedure, approved by COMPANY.”

  With: GMAW automatic welding shall be additionally fully inspected in accordance with an Automatic Ultrasonic Test (AUT) procedure, approved by COMPANY.

- Replace sub paragraph e): ”Charpy Impact test sets in accordance with ASTM A370 shall be taken from the 12 o’ clock, 3 o’clock and 6 o’clock positions as follows:”

  With: Charpy Impact test sets in accordance with ASTM A370 shall be taken from the 12 “O” clock, 3 “O” clock, 6 “O” clock and 9 “O” clock positions as follows:

- Replace the entire sub paragraph h) (i) and h) (ii),

  With:

  h) (i) Crack tip opening displacement (CTOD) tests shall be conducted at 0 degree C or minimum design temperature, whichever is lower, for all welding procedure qualification tests (except Fillet welds) of Gas pipelines with pipes having thickness of 12.5 mm (½ inch) and above to ensure integrity of girth welds.

  h) (ii) As the fracture toughness shall be regarded as size sensitive, qualified for the thickness tested, the CTOD values shall include thickness tested. The tests shall be conducted in accordance to BS EN ISO 12737 and BS EN ISO 15653:2010 using SENB (Single Edge Notched Bend) geometry test specimens.

- Replace sub paragraph h) (iii): “For each welding procedure, both the weld metal and Heat affected Zone (HAZ) at 12 “O” clock, 3 or 9 “O” clock and 6 “O” clock positions shall be tested as per API 1104, Appendix A, Section A.3.2.3.3.”

  With: For each welding procedure, both the weld metal and Heat affected Zone (HAZ) at 12 “O” clock, 3 or 9 “O” clock and 6 “O” clock positions shall be tested as per API 1104 latest edition, Appendix A, Section A.3.4.3.

- Add following at the end of sub section 12.5:

  COMPANY acceptance is required before sectioning weld qualification tests coupon for mechanical testing. EPCm CONSTRUCTION CONTRACTOR also to ensure the final cut/machined test coupon is not affected by heat from the cutting/machining operation.

  All mechanical testing shall be performed by an ISO 17025 certified testing laboratory approved by COMPANY.

  If any mechanical test specimen fails to meet the test requirements, two sets of retests, for that particular type of test, may be performed upon COMPANY approval. The result of both retest specimens shall meet the specified requirements. All the results shall be reported and be a part of the WPQR package.

4.8 Amendments to Section 13.0 – Welder Qualification

Sub Section 13.1 Qualifications of Welders

- Replace sentence in the first paragraph: "Testing shall be in accordance with API 1104 Section 3."

  With: Testing shall be performed in accordance with Section 6 and Section 10.4 of API 1104 latest edition for new welds and repair welds, respectively. A new PROJECT specific welder and welding operator qualification programme shall be established, and previous/existing qualifications are not acceptable.

- Replace sentence in the last paragraph: "At the discretion of COMPANY, the welder in question may be retested after an agreed upon period of retraining."

  With: At the discretion of COMPANY, the welder in question may be retested after an agreed period of retraining.

4.9 Amendments to Section 14.0 – Production Welding

Sub section 14.1 General Requirements

- Add following at the end of sub section 14.1:

  Production welding shall not commence until all welding documentation package has been approved by COMPANY.

  Firm supports shall be provided for the pipes during welding operation. The pipes shall not be hold using boom truck crane slings or similar equipment, as it could potentially cause movement leading to distortion/crack on the weldments.

  Prior to welding the girth welds by automatic welding, EPCm CONSTRUCTION CONTRACTOR to scribe a reference line on the surface of pipe on ether sides of the welds for the AUT examination scanner band width. The distances shall be in accordance with the approved AUT procedure.

Sub Section 14.4 Weather Limitations

- Add following at the end of sub section 14.4:

  Adequate ventilation shall also be provided to control the quality of breathing air during the welding operation.

Sub Section 14.5 Preheating and Interpass Temperature and Post Weld Heat Treatment

- Add following at the end of sub section 14.5:

  Temperature indicating crayons, thermocouples, non-contact infrared thermometer or contact pyrometers shall be used to monitor preheat and interpass temperatures. Temperature indicating crayons is not permitted to be used for joint that is to be coated or that will have a heat-shrink sleeve installation on it.

The temperature of the weld area shall be monitored throughout welding. The temperature shall not be allowed to drop below the minimum preheat temperature and not be higher than the maximum interpass temperature established during welding procedure qualification.

Sub Section 14.9 Arc Strikes

- Add following at the end of sub section 14.9: EPCm CONSTRUCTION CONTRACTOR shall notify COMPANY in the case stray arcs occurs outside the weld area of the bevel. The area shall be removed by grinding and subject to magnetic particle testing to ensure freedom of cracks and the remaining wall thickness shall also be checked by ultrasonic inspection. The pipe shall not further be used unless accepted by COMPANY.

Sub Section 14.11 Marking Welds

- Add following at the end of sub section 14.11: Paint or marker to be used for identification shall be free from sulphur, chloride and other halogens.

4.10 Amendments to Section 15.0 – Standard of Acceptability

- Replace: “The standards of acceptability for all welding shall be in accordance with the requirements of API Standard 1104, Section 6, and this Specification.”

  With: The standards of acceptability for all welding shall be in accordance with the requirements of Section 9 of API Standard 1104 latest edition and this addendum.

Sub Section 15.1 – Inspection and Testing – General Requirements

- Add following at the end of sub section 15.1:

  Additional inspection of any weld joint at any stage of production welding may be requested by COMPANY, including re-inspection of previously inspected joints. In the case such testing disclose non-conformance to the acceptance criteria, all repair and NDT costs shall be done at the EPCm CONSTRUCTION CONTRACTOR’s expense.

  Preparation of NDE procedures, specifications, and work instructions shall be carried out under the responsibility of NDE level III, and clearly state the approval of the NDE level III.

  All automatic welds shall be inspected by auto UT (AUT) based on criteria specified in API1104.

  Radiography is not warranted where auto UT is carried out. However, for verification purpose, first 20 number of automatic weld joints shall be subjected to both AUT and Xray for comparison purpose.

  COMPANY requires procedure approval, calibration and validation for both AUT and PAUT. AUT shall be calibrated based on API 1104. PAUT shall be calibrated based on Code CASE 181 of ASME B31.3.

  AUT Consultant in addition, EPCm CONSTRUCTION CONTRACTOR shall assign ASNT

level III in UT/AUT through Independent NDE agency for full time witness and interpretation of the results at site.

Sub Section 15.3 – Ultrasonic Inspection

- Replace: ”Excluding (iv) and (v), acceptance shall be in accordance with the relevant pipe material specification.”

  With: Excluding d) and e), acceptance shall be in accordance with the relevant pipe material specification.

4.11 Amendments to Section 16.0 – Repair of Welds

- Replace: ”The decision to proceed with repair in each instance shall be approved by COMPANY. Only qualified, COMPANY approved repair procedures shall be used.”

  With: The decision to proceed with repair in each instance shall be approved by COMPANY. Only COMPANY approved and qualified repair procedures shall be used. A repair procedure shall include method statement regarding excavation, NDT after excavation, and repair welding requirements.

- Add following at the end of sub section 14.9: Weld repairs using Low Hydrogen electrodes are only permitted, the use of cellulosic coated welding electrodes is not permitted on weld repairs.

4.12 Amendments to Attachment 1 – Vickers Hardness Measurement Traverses (Figures)

- Replace: Figure 1

  With: Figure 2 & 3 (Para 7.3.3.3) of NACE MR0175/ISO 15156-2:2020

4.13 Add New Section: NON-DESTRUCTIVE TESTING

4.13.1 GENERAL

This section covers requirements for non-destructive examination of field or site welded pipeline girth and tie-in welds including fillet welds.

Non-destructive Testing (NDT) procedures shall be developed and qualified in accordance with API 1104 and this specification & its attachment.

NDT Plan, Procedures and Qualification data shall be submitted to COMPANY for review and approval prior to the start of welding.

EPCm CONSTRUCTION CONTRACTOR shall furnish the services of a qualified, experienced NDT testing company registered in the commercial directory. Such company shall provide testing and written signed reports of all findings.

All the NDT Equipment shall have a Valid Calibration during the testing and Calibration Certificates shall be submitted.

To avoid missing out of any delayed cracking, all the NDT shall be performed after 24 Hrs

after completion of the welding.

The NDT acceptance criteria shall be in accordance with Section 9 of API 1104 and this specification & its attachment.

When visual or non-destructive inspection demonstrates that a weld fails to comply with the project requirements, the defective weld shall be repaired and re-examined. All repairs shall be carried out in accordance with API 1104 and this specification. Only qualified, COMPANY approved repair procedures shall be used.

Surface preparation / coating shall be commenced only after acceptance of the applicable NDTs.

4.13.2 NDT PERSONNEL

The qualifications and certifications of all NDT personnel shall be according to ASNT SNT-TC-1A or COMPANY approved equivalent standards. In addition, NDT Personnel shall meet Vision Examinations requirement of Clause 8.4.3 of API 1104. At least one ASNT Level III Examiner shall be assigned to the project. EPCm CONSTRUCTION CONTRACTOR shall assign ASNT level III in PAUT / AUT / RT through Independent NDE agency for full time witness and interpretation of the results at site/certification.

COMPANY shall approve the NDT testing company utilized by Construction Contractor. All NDT Personnel shall have a minimum 5 years' experience on pipeline weld testing. The qualifications of all NDT personnel used on the project shall be submitted for COMPANY approval.

Only Level II or III personnel authorized for calibration of equipment and test results interpretations.

All welding inspectors shall be American Welding Society Certified Welding Inspectors. EPCm CONSTRUCTION CONTRACTOR AUT personnel shall have attended factory training for the specific Automated Ultrasonic Testing System being used for Project inspection.

EPCm CONSTRUCTION CONTRACTOR shall submit all required personnel qualification and certification records with bid.

4.13.3 VISUAL INSPECTION

All welds shall be subjected to 100 percent visual inspection. Acceptance criteria shall be according to API 1104.

The weld surface is sufficiently free from coarse ripples, grooves, overlaps, and abrupt ridges and valleys. The surface condition of the finished weld shall be suitable for proper interpretation of radiographic and other required non-destructive examinations.

The inside diameter weld reinforcement shall not be raised above the parent material by more than 1/16 in. (2 mm). The OD weld reinforcement shall not be raised above the parent material by more than 1/8 in. (3 mm), at no point shall it fall below the outside surface of the

pipe.

In addition to API 1104 requirements, filler wire protruding into the pipe bore is not permitted.

4.13.4 MAGNETIC PARTICLE TESTING

Magnetic Particle Testing (MT) shall be performed according to a written procedure conforming to API 1104 and this specification & its attachment. All welds originally inspected by Automated Ultrasonic Inspection and all fillet welds shall be inspected by MT.

The prod method as per ASME Section V, Article 7, T-743 shall not be used.

Either of the following methods shall be used for the completed welds (e.g., after all welding, heat treatment, and grinding) where MT inspection is specified.

- The white contrast, red particle (wet) method.
- The yellow contrast, black particle (wet) method.

MT consumable in form of ready to use pre-mixed Aerosol Cans shall only be used or any alternate methods needs to be approved by the COMPANY.

Residual magnetism shall not exceed 2.5mT or 25 Gauss after the MT. If the residual magnetism found to be more, Demagnetization shall be done to remove the residual magnetism after completion of the test. Residual magnetism shall be measured for each joint and the same shall be stated in the ITP.

Calibration blocks and approved magnetic flux indicators shall be always available on Site.

Post cleaning is mandatory for all surfaces including, white paint if used to enhance the color contrast, it shall be removed as soon as practical using a process that does not adversely affect the part.

Acceptance criteria shall be in accordance with the requirements of API 1104 and this specification.

4.13.5 RADIOGRAPHIC TESTING

Radiographic Testing (RT) shall be performed according to DGS-6300-005 and API 1104 and this specification & its attachment. EPCm CONSTRUCTION CONTRACTOR shall submit RT procedures at least 60 days prior to start of construction and qualify those procedures at least 30 days prior to start of construction.

EPCm CONSTRUCTION CONTRACTORS using radiographic testing shall be equipped with X-ray crawler for main pipeline assembly inspection. Single wall exposures shall be used when possible. Double wall exposures may be used for tie-in welds and repairs.

Fine-grained film and lead intensification screens shall be required as per DGS-6300-005 (Refer Appendix-2).

The Zero datum shall be located at the 12 O'clock (top) position. The direction of numbering on tape shall be clockwise when looking in the direction of flow. Defect location shall be in

the metric system. Each radiograph shall have reference markers at 100 mm (minimum) intervals.

The overlap for the film shall not be less than 25mm. The minimum width of the film shall cover full weld toe to toe plus 25 mm either side of the weld. For repair welds, the film shall cover 100 mm on either side of the weld repair area.

Acceptance criteria shall be in accordance with the requirements of API 1104 and this specification & its attachment.

Radiographs shall be retained by EPCm CONSTRUCTION CONTRACTOR as per DGS-6300-005. Radiographs shall be stored by EPCm CONSTRUCTION CONTRACTOR in environmentally controlled, fire-resistant locations. Radiographs shall be available for COMPANY inspection upon request.

4.13.6 MANUAL ULTRASONIC TESTING

Manual Ultrasonic Testing (MUT) of welds shall be performed according to a written procedure conforming to API 1104 or an equivalent standard approved by Company.

MUT shall be performed by EPCm CONSTRUCTION CONTRACTOR on all AUT repairs and any AUT result requested by COMPANY.

Acceptance criteria shall be in accordance with the requirements of API 1104 and this specification & its attachment.

4.13.7 AUTOMATED ULTRASONIC TESTING (AUT)

4.13.7.1 ***General***

AUT shall be used as the primary non-destructive inspection method for pipeline construction on the Project. In those cases where AUT is not technically suitable, RT shall be applied. All pipeline welds shall be inspected by either AUT or RT over their full length. If AUT is applied in lieu of Radiography for pipeline girth welds welded by automatic GMAW, for verification purpose, first 20 number of automatic weld joints for each NDE Sub contractor / spread, shall be subjected to both AUT and X-Ray for comparison purpose.

Automated Ultrasonic Testing (AUT) of welds shall be performed according to a written procedure conforming to API 1104, this specification and ASTM E 1961. EPCm CONSTRUCTION CONTRACTOR shall develop specific technique sheets for pipe diameter, bevel design, and wall thickness to be inspected for each NDE Sub contractor / spread. Both the AUT Procedure and all specific technique sheets shall be submitted to COMPANY for approval.

The inspection shall be performed with a mechanized system that utilizes phased array, pulse-echo, and/or pitch-catch focused transducers with zoned inspections and Time of Flight Diffraction (TOFD) techniques. The system shall create and retain an electronic record of the ultrasonic inspection results for the duration of the pipeline construction. The electronic record shall be submitted to the COMPANY at completion of the project.

EPCm CONSTRUCTION CONTRACTOR shall demonstrate the AUT system ability to accurately size length and vertical height of indications with a resolution compatible with the applicable acceptance criteria. This shall be demonstrated on calibration blocks with machined notches and holes per ASTM E 1961. Calibration blocks shall be prepared from project pipe in each pipe diameter, pipe thickness to be inspected. Each AUT work crew shall have its own dedicated set of calibration blocks for the inspection being performed.

Acceptance criteria shall be in accordance with the requirements of API 1104 and this specification &its attachment.

AUT shall be supplemented with MT inspection for near surface defects. AUT system shall be calibrated at the beginning of each work shift and after the inspection of each ten welds. AUT system shall be re-calibrated upon COMPANY request.

AUT system shall electronically record circumferential weld distance and any indications within 1 cm accuracy over the entire weld circumference. AUT technicians shall maintain a System Logbook detailing the performance characteristics data and identification for instruments, transducers, and inspection set-ups. Register of all Calibration Blocks including unique number shall be included in System Logbook. All changes in components or procedures shall be documented in the Logbook which shall be available for COMPANY inspection on request.

AUT system shall be qualified, calibrated, and operated in accordance with the referenced standards of this specification. The AUT qualification program shall establish the inspection accuracy from a statistically significant number of welds. Test welds shall contain a sufficient number of flaws to establish an 85% confidence of 95% accuracy.

AUT inspection results shall be provided to COMPANY Representative and Construction Contractor within 4 hours of completion of each weld. AUT Daily Inspection Report shall be issued by 10 am following day to include:

- Equipment identification
- Pipeline identification and kilometer location
- Weld identification
- Datum location
- Length of defects
- Depth from O.D. surface to bottom of defect
- Height of defect
- Defect classification (linear, transverse, volumetric)
- Examination personnel identity and qualification level Date examination performed.

4.13.7.2 ***Equipment***

The inspection shall be performed with a mechanized system that utilizes phased array, pulse-echo, and/or pitch-catch focused transducers with zoned inspections and Time of Flight Diffraction (TOFD) techniques. The sound beams may be generated by phased array transducer along with conventional transducers. The system shall provide an adequate number of inspection channels and the inspection shall be designed with sufficient beam overlap to ensure the complete volumetric examination of the weld throughout the thickness in one circumferential scan.

The system incorporates an integral recording system to indicate the location of imperfections and also a system to monitor the acoustic coupling of the phased array probes. The instrument shall provide a linear “A” scan presentation for each channel selected, as well as inspection channels to allow the volume of the weld scanned, to be assessed in accordance with the defined inspection zones.

The system shall demonstrate the ability to accurately size length and vertical height of indications with a resolution compatible with the supplied acceptance criteria. Indication length is measured using the system distance encoder. Encoder accuracy can be checked by encoded distance recorded versus physical measurement on the pipe.

Transducers used for Pulse echo, TOFD or phased array shall be approved by the COMPANY with regards to type, Sizes, beam angles, frequencies, contour etc.

The ultrasonic system shall include scanner heads and system set-up specifically configured for testing of repairs where the primary function is to confirm the complete removal of rejected defect.

All AUT Equipment shall have valid calibration (with 12 months) the equipment shall be recalibrated if any change of parts due to malfunctioning or a deviation from the expected performance is noticed during the normal operations.

AUT equipment shall be in line with project specifications & ASTM E-1961 & linearity checks shall be performed for every 6 months. Instrument linearity shall be such that the accuracy of any indicated amplitude is within 5% of the actual full-scale amplitude.

All recording, data handling and presentation software shall be identifiable by a unique version number for AUT system. The software version number in use, shall be clearly observable on all display and printout presentations of calibration and examination results. Any changes of software versions shall be avoided during a project. Any necessary changes shall be approved by the COMPANY.

4.13.7.3 Calibration / Reference Blocks

4.13.7.3.1 General

Calibration blocks shall be used to set AUT system sensitivity, and to verify the inspection system for field inspection and to monitor the ongoing system performance. Calibration blocks shall be manufactured from a section of pipeline specific line pipe. The wall thickness

of the pipe used for calibration blocks shall correspond to the nominal wall thickness of the pipes used. No weld seam allowed on the calibration block.

Acoustic velocity and attenuation measurements shall be performed on material from all sources of pipe material supply to be used. If differences in acoustic velocity for the same nominal wall thickness from any source of supply results more than +/-100 m/sec, specific calibration blocks shall be made for material from each source of supply showing such variations.

Details of the specific weld bevel geometries including relevant dimensions and tolerances shall be provided in order to determine the particulars and numbers of calibration blocks required.

Calibration block records shall include: Ultrasonic certificate and Machining certificate.

The calibration standard and mounting shall be equipped with a permanently mounted welding band to allow the array to be correctly positioned and moved during dynamic calibration scanning.

4.13.7.3.2 Acoustic Properties

Acoustic velocity and attenuation measurements shall be performed on material from all sources of pipe material supply to be used. These measurements shall be performed according to ASTM E 1961.

4.13.7.3.3 Dimensions

Overall dimensions of the calibration standard and its actual shape shall be determined by the size of the transducer array and block support structure. The standard (and its mounting) shall provide sufficient surface area to permit the entire transducer array to traverse the target area of the calibration standard during dynamic calibration scanning. Details of the calibration standard dimensions are included in the Cal Block documents for each thickness.

Dimensional verification of all calibration reflectors and their position shall be performed and recorded according to a documented procedure. Each reflector shall be verified by ultrasonic testing to ensure that the reference reflector pairs yield reflective signals within +/-3dB.

The machining tolerances for calibration reflectors are:

(a) Hole diameters •+/- 0.2 mm

(b) Flatness of FBH •+/-0.1 mm

(c) All pertinent angles •+/- 1°

(d) Notch depth •+/- 0.1 mm

(e) Notch length •+/- 0.5 mm

(f) Central position of reference reflectors •+/- 0.1 mm

(g) Hole depth •+/- 0.2 mm

4.13.7.3.4 Identification

All standards shall be permanently identified with a unique serial number.

Information recorded for each calibration standard shall include; diameter, wall thickness, and serial number. The serial number shall be used to trace recorded information with regards to acoustic velocities, manufacturer, and target information.

A register of all calibration blocks and records shall be established including the unique serial number, wall thickness, bevel design, diameter, and ultrasonic velocity for each calibration block and shall be always maintained and available for COMPANY review.

4.13.7.3.5 Targets

EPCm CONSTRUCTION CONTRACTOR shall indicate Calibration settings and Tolerances in the procedure including for TOFD lateral wave set in accordance with the following Table:

Channel Name	Set Value (%FSH)	Minimum Value (%FSH)	Maximum Value (%FSH)
Root	80 (1 mm Notch)	64	100
Root Map	80 (1 mm Notch)	64	100
Fill	80 (2 mm FBH)	64	100
Cap	80 (1 mm Notch)	64	100
LCP	80 (2 mm FBH)	64	100
HP1/HP2	80 (2 mm FBH)	64	100
Volumetric	80 (1.5 mm FBH)	64	100
Transverse ID/OD	80 (1 mm Notch)	64	100
TOFD*	40% FSH Lateral wave	40% FSH Lateral wave	40% FSH Lateral wave
TOFD ID* (multigroup)	80% FSH Backwall	64% FSH Backwall	100% FSH Backwall
*TOFD lateral wave set, minimum and maximum sensitivity levels shall be utilized as an inspection guideline.			

4.13.7.3.6 Thickness Survey

A thickness survey shall be performed on the calibration block material prior to machining. The survey shall consist of manual scan of the block, in perpendicular strips with no more than 20 mm separation.

A compression wave transducer, with nominal frequency of at least 5 MHz, with crystal diameter of 6 to 10mm, should be utilized. Maximum variation in thickness (min. / max.) for each strip scan shall be recorded and included as part of the calibration block records.

4.13.7.3.7 Manual Ultrasonic Scan

For each reflector in the calibration block, its positional response shall be verified with a manual ultrasonic scan.

1. Using the same angle transducer for each reflector that will be used for production testing, the distance from the weld centerline at maximum response shall be determined and recorded.
2. At the maximum response location, the dB gain to provide 80% FSH from the reflector shall be determined and recorded.
3. For any distance to centerline measurements that vary more than ±1.0 mm between each opposing pair of transducers, the cause of the variation shall be determined. Any deviation outside of tolerance shall be cause for recorded corrective action.
4. For any gain response that varies more than ±3 dB between each opposing pair of transducers, the cause of the variation shall be determined. Any deviation outside of tolerance shall be cause for recorded corrective action.

4.13.7.3.8 Calibration Block Acceptance and Usage

For any calibration block whose parameters exceed those defined above, the calibration block shall be deemed unacceptable. An acceptable remedy shall be carried out or a new block shall be manufactured.

Each AUT work crew shall have its own dedicated set of calibration blocks for the inspection being performed.

AUT system shall be calibrated at the beginning of each work shift and after the inspection of each ten welds.

AUT system shall be re-calibrated upon Company request.

4.13.7.3.9 AUT Procedure Requirements

EPCm CONSTRUCTION CONTRACTOR shall develop Project specific procedure, AUT technique sheets shall be developed for each pipe diameter, bevel design and wall thickness to be inspected. Both the AUT Procedure and technique sheets shall be submitted to Company for approval.

The system shall demonstrate 100% scanning of the weld and heat affect zone (HAZ) with adequate beam coverage overlap and signal strength for relevant imperfection sizes.

AUT system shall be capable to characterize the defect / indication, oriented longitudinally, transversely or across the thickness in terms of length, width and depth with the required resolution.

A typical set-up for AUT inspection of girth-welds shall be a combination of:

- Tandem mode inspection for the inspection of fusion line on upstream and downstream sides either using phased array probes or multi-probes systems.
- Pulse-echo inspection for root and cap
- and TOFD

UT beams shall be focused on each zone and sizing is amplitude based (which means that a reference: either a flat bottom hole or a notch is associated to each zone). Beam focusing is adjusted so that the overtrace in adjacent zones must typically be between -6 and -10dB.

Pipe Wall thickness variations must be measured. If wall thickness variations exceed 1.5 mm, more than one inspection set-up and multiple calibration blocks must be used. In production wall thickness must be measured continuously to allow using the relevant set-up for the actual wall thickness of the inspected area.

The maximum acceptable offset for Scanner positioning must be qualified. Typical value is ± 1.5 mm or ± 2.0 mm for probes tolerance positioning. Weld shrinkage must be measured to position the probes with taking into account the effect of shrinkage on fusion lines positions.

In case of repair, AUT can only be used to check that the original defect has been removed in addition, manual UT and RT shall be performed in the repair area.

As a minimum the AUT system with its normal set-up, but with gates wide enough to encompass the repair area and confirm that the AUT defects have been removed. During this special attention shall also be made to TOFD channel indications, and indications outside the normal gate settings.

The information provided by all AUT channel types shall be actively used in order to ensure adequate imperfection detection and sizing.

The primary requirement to any AUT system is that its performance is documented in terms of adequate detection and sizing, or rejection abilities, in relation to specified/determined acceptable imperfections. The performance of the AUT system shall be through the qualification to be demonstrated to meet or exceed the requirements in terms of detection or rejection set by the applicable acceptance criteria specified in this document.

Recording thresholds should be selectable to display signals between 0% and 100% of full screen height for simple amplitude and transit time recording and should be from 0% to 100% for B-Scan or “mapping” type recording data. Two recordable signal outputs per gate should be available, being either analog or digital and representative of signal height and time of flight. These will be suitable for recording on a multichannel recorder or computer data acquisition software display” “B-Scan or “mapping” display will be used for volumetric flaw detection and characterizations and Time of Flight Diffraction (TOFD) technique may be added to improve characterization and sizing. TOFD techniques may augment pulse-echo techniques but shall not replace pulse-echo techniques.

4.13.7.3.10 AUT Procedure Validations

Project specific AUT procedure validation shall be performed in order to demonstrate that, the project specific AUT procedure can consistently accomplish the required detection capabilities prior to its application on inspection of production joints.

Project specific validation process shall be witnessed and approved by a competent independent 3rd party.

EPCm CONSTRUCTION CONTRACTOR shall submit a detailed project specific validation procedure which is duly approved by the 3rd Party AUT specialist for COMPANY review and approval.

Essential variables for validation of AUT technique: Base Material, Diameter, thickness, grove angle, Welding process, WPS, Welding Consumables, AUT system including weld ring, transducer setup, Reference reflectors, working temperature range, data acquisition and Software, travel speed dynamics, method of scanning (outside only or both inside and outside of the pipe), direction of scanning etc.

Defects in Test Coupons shall be created by degradation of the welding process, welding parameters instead of machined or manually induced defects, as much as possible. The induced defects shall vary in length, height and location. Too close spacing of defects shall be avoided. All test coupons shall be subjected to Radiography testing using Ultra fine grain film and Magnetic particle testing as a supplementary NDT.

Scanning shall be performed on both upstream and downstream sides of the Test coupon to avoid masking the near side defect.

Qualification testing shall include:

- Repeatability test program (Including band offset tests)
- Temperature Sensitivity test program
- Reliability test program

4.13.7.3.11 Repeatability Test Program

This program is to check the system repeatability and stability, based on calibration scans and location on the pipe. The maximum deviations for each calibration block position will be considered. A maximum deviation outside ±2dB is not acceptable. For band offset scans made on deliberate defect weld, the criteria shall be that the detection and sizing tolerance shall be equal to the nominal values obtained from optimum scans.

4.13.7.3.12 Temperature Sensitivity Test Program

Temperature affects the sound path in the wedge (deviation of the UT beam), the acceptable temperature range for inspection must be qualified at lowest and highest possible ambient temperatures as per UAE's meteorological data. A maximum deviation outside ±2dB is not acceptable.

4.13.7.3.13 Reliability Test Program

Reliability tests shall be subjected to Destructive Reference testing by weld sectioning to determine the position of the indication / defect (PoD) as a function of defect height and size of the defect.

All relevant observations in reliability tests shall be plotted as AUT defined height versus measured height from the validation testing, for evaluation of defect height sizing accuracy. The plots shall include a line representing the 5% fractile against under-sizing (i.e., less than

95% probability of under-sizing) for each defect type and depth location.

A probability of rejection (POR) analysis should be performed, as a validation of PoD and height sizing accuracy.

Sizing of the defect shall be against Salami cross sectioning. Number/extent of sections for each position (Root, Hot pass, Fill, Cap etc.) shall at least include all deliberate flaws included in section C (Table) in addition, a random weld area with no indication from AUT or supplementary NDT shall also be selected for sectioning. The defect location, height and depth shall be measured in millimeters with an accuracy better than ± 0.1mm.

4.13.7.4 Calibration and System Compliance Check

4.13.7.4.1 Calibration Scan and Adjustment

A calibration scan shall be the official record for sensitivity and gate positioning.

Additional calibration scans will be required if the initial calibration scan demonstrates sensitivity outside the defined tolerance. If a subsequent calibration falls within the tolerance of the calibration, it shall be considered successful only if no changes to instrument settings are performed.

In the event that adjustment is required, adjustment will be applied, and additional calibration(s) will be performed.

The TOFD display on the calibration scan must demonstrate the ability to detect the smallest applicable root notch with a response at least 6 dB greater than adjacent grain noise. If the applicable TOFD root notch is not discernible in the calibration scan, the calibration is considered failed.

The lateral wave in the TOFD display should have an adequate response to detect the OD TOFD notch.

All inspection gates shall be set to ensure complete coverage of the weld volume.

Settings of the gates shall be verified by the dynamic calibration scan.

The gate lengths for the volumetric channels must be sufficiently long that the data visualization shall provide evidence of the root and cap re-enforcements.

The calibration scan shall be conducted at the same scanner speed as the weld inspection scan and with the same couplant.

4.13.7.4.2 Calibration Intervals

The system shall be Calibrated by scanning the calibration block as follows.

The calibration of the system shall be verified by scanning the calibration block before and after inspection of each weld for the First 20 welds & thereafter for every 10 production welds. The gain added to any channel for field examinations shall be removed during verification scans.

The following shall be reported in the same working shift.

a) At any change of calibration block.

b) At any change of nominal wall thickness.

c) At any change of components.

d) Start and end of each day / shift.

e) Any change resulting in the change of beam angle +/- 2° in steel.

f) Any change in the frequency of probe or any probe parameter.

g) Any change in the material composition, diameter and thickness.

h) If any calibration during production is shown to be below acceptable sensitivity, all welds scanned since the previous good calibration shall be re-scanned to ensure compliance. Calibration is then required before or after each of the following 20 welds before reverting to the lower frequency.

4.13.7.4.3 System Compliance Checks

To ensure the system remains in compliance with its set-up settings and mechanical parameters

Periodic verification of the system shall be carried out. These shall include the following:

1. A visual check that the wear pins on the probe shoes are not worn down and are set at the required Distance of between 0.2 and 0.5 mm from the shoe face.
2. The adjustment of any mechanical fastening that could affect the probe separation or scanner height
3. Shall require a calibration verification scan prior to re-use. General maintenance e.g. Oiling of Bearings, re-greasing of motors or adjustment to nut, bolts etc. that do not entail adjustments to the Probe assembly does not require calibration verification scans.
4. Visual check on the calibration block to verify that any grooves caused by the wear pins do not Exceed 0.5 mm in depth. Suitable equipment shall be available to measure the depth of the grooves Caused by the wear pins.
5. A visual check on the umbilical cable to check for wear and tear. Remedial action shall be carried out as required to ensure that the cable is not degraded to a point where it becomes unusable.
6. The adjustment or replacement of any mechanical fastening or probe holder part that could affect the Probe separation or scanner height in any way, or the replacement of a probe pin, umbilical cable or Probes shall require a calibration verification scan and adjustments as necessary prior to use.
7. Prior to commencement of testing and at periodical intervals thereafter the circumferential position Accuracy shall be verified to ensure it remains within the

permitted 10 mm tolerance.

8. A number of verification and integrity checks shall be carried out prior to commencement of any work or at any time a potential problem is encountered, these include but are not restricted to the following:
   - Cable integrity
   - Data acquisition unit integrity
   - Transducer element verification
   - Scanner interface module verification

4.13.7.5 Weld Inspection

4.13.7.5.1 Surface Condition

A 150mm (~6”) wide scanning area on each side of the weld in OD surface and 75mm (3’’) on each side of the weld in ID surface shall be clear of weld spatter, coating, and other conditions which may interfere with the movement of the transducers, the coupling liquid, or the transmission of acoustic energy into the material. The surface condition on the OD surface may be achieved by power brushing and grinding.

Long seams shall be ground flush on the OD and ID surface for sufficient distance to allow scanning. Any surface condition such as geometry, coating, etc., impeding the ultrasonic inspection shall be noted for corrective action prior to scanning.

3LPE coatings shall be removed 150mm from either side of the pipe edges to allow scanning. Pipe bare surface shall be available for band set and scanner clearance.

4.13.7.5.2 Scribe Line

Following the weld beveling process, prior to pipe fit-up, a reference line will be scribed on the pipe surface, over the entire circumference, on one end of each pipe whenever possible. The reference line will be set from the weld center line and will be utilized during scanner band placement and alignment process, to ensure probes remain at the correct distance from weld centerline for the complete weld circumference. For each varying weld configuration, should be scribed on both ends to enable weld shrinkage measurement to be calculated.

Shrinkage will be accounted for during scanner band placement.

The line should be heavy enough to remain visible after post weld cleaning but not heavy enough to be considered detrimental to the integrity of the pipe. In the event that a scribe line is not present, the band shall be set to the visual center of the cap pass. The band may need further repositioning and associated re-scans to ensure the root and cap geometries are centered.

4.13.7.5.3 Scanner Guide Band

The scanner guide band will be placed on pipe at a fixed distance from scribed line to ensure probes remain at the correct distance from weld centerline (compared to the dynamic

calibration scan), during scanning process, and for the complete weld circumference. The guide band shall be positioned at the defined offset distance from the weld centerline with an accuracy of ± 1.0mm (based on the scribe line).

4.13.7.5.4 Zero Datum Point & Scan Direction Identification

The zero-datum point is marked at the 12 O'clock position and scan direction (clockwise or counterclockwise) is signified by an arrow; both will be marked with a permanent marker or equivalent. AUT downstream is considered to be the side of the weld where the band is mounted.

4.13.7.5.5 Surface Temperature

The surface temperature of the weld under examination shall be withing the range qualified in procedure validation. The temperature measurements shall be taken by digital laser temperature gauge.

4.13.7.5.6 Weld Identification

Each weld shall be numbered in the sequence used in the supplied pipe tracking system. The weld numbering system shall be provided prior to any production weld testing & it should be documented along with the report and the numbering system shall be incorporated in the weld/NDE summary.

4.13.7.5.7 Weld Scanning and Evaluation

The following scanning and evaluation threshold guideline shall be utilized for scanning:

Channel Name	Reference Reflector	Display Mode	Reference Sensitivity Level	Scanned Sensitivity Level	Evaluation Threshold
Cap	1 mm Notch	Pulse Echo	80% FSH	Reference	20% FSH
Fill	2 mm FBH	Pulse Echo	80% FSH	Reference	40% FSH
LCP	2 mm FBH	Pulse Echo	80% FSH	Reference	40% FSH
HP1/HP2	2 mm FBH	Pulse Echo	80% FSH	Reference	40% FSH
Root	1 mm Notch	Pulse Echo	80% FSH	Reference	20% FSH
Root Map	1 mm Notch	Mapping	80% FSH	+4dB	40% FSH
Volumetric	1.5 FBH	Mapping	80% FSH	+6dB	40% FSH
Transverse ID/OD	1 mm ID+OD Notch	Mapping	80% FSH	Reference	40% FSH
TOFD	ID+OD TOFD Notches	B-Scan / Grey scale	40% - 80% LW	Reference	Baseline
TOFD ID (multigroup)	ID+OD TOFD Notches	B-Scan / Grey scale	80% FSH BW +10dB	Reference	Baseline
Coupling Channels	Backwall	Go / No Go	40% FSH	40% FSH + 10dB	40% FSH

Indications caused by root or cap reinforcement geometry do not require evaluation or

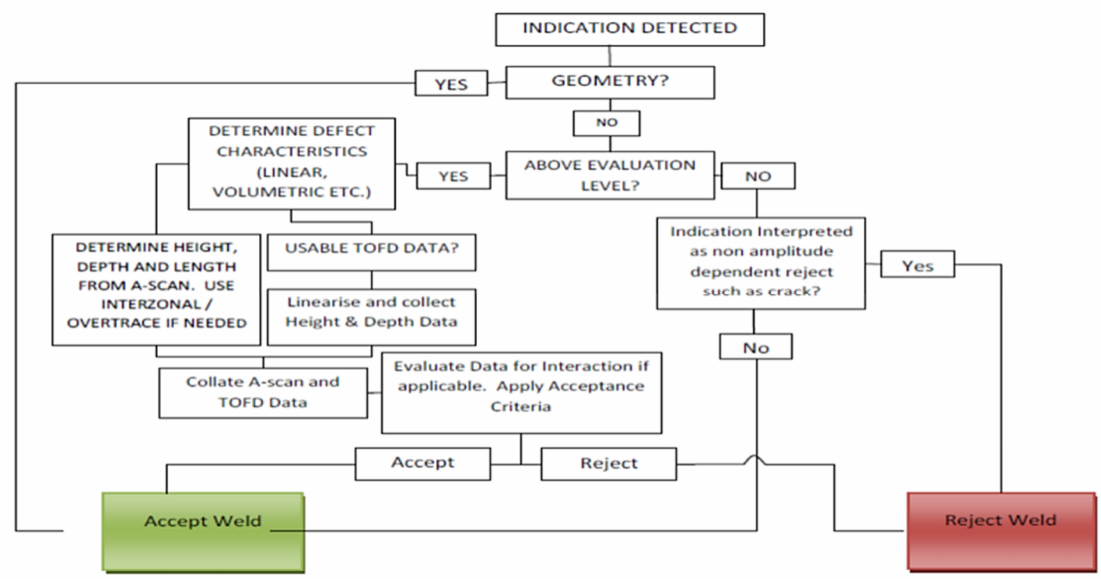
reporting.

In any case of doubt with geometrical indications it should be verified by visual & MPI methods. If required ground flush should be performed before MPI.

Indications recorded from sources other than weld imperfections exceeding length tolerances shall be evaluated. The nature of such indications shall be determined.

All indications shall be reported which ever crossed the reference level which are near to the acceptance level.

Flowchart enhances the evaluation for acceptance and rejection of weld subjected to finally review by the COMPANY.



4.13.7.6 Acceptance Criteria

4.13.7.6.1 Linear Indications Threshold Limits

Linear Indications threshold limits shall be as described below.

1. The maximum acceptable length of buried defects is the lesser of the following:
   - 50 mm (2 in.) in total length in a continuous 305 mm (12 in.) length of weld.
   - 8 percent of the total weld length if the amplitude in one or more fill channels is greater than 20 percent or if the amplitude in one or more Porosity channels exceeds 50 percent (cold lap indication). Defect interaction is not considered in length evaluation. Defect length shall be corrected for beam spread. For length determination the background level is defined to be 20 percent for the Fill and VOL channels.
2. The maximum acceptable aggregate length of defects in the Root or Cap channels interpreted to be open to the surface is the lesser of the following:

- 25 mm (1 in.) in a continuous 305 mm (12 in.) length of weld.
- 8 percent of the weld length.

For the Root and Cap notch channels, the threshold is 40 percent based on the 1.0 mm (0.040 in.) notch calibration. Defect interaction is not considered in length evaluation Defect length shall be corrected for beam spread. For length determination, the background levels for the Root and Cap notch channels are defined to be 20 percent amplitude.

3. If the top Fill channel is used to inspect the weld OD and if at least one amplitude value is greater than or equal to 20 percent, then indicating a surface defect, the acceptable aggregate length in a continuous 302 mm (12 in.) length of weld is the lesser of the following:
   - 25 mm (1 in.)
   - 8 percent of the weld length

Defect length shall be corrected for beam spread. For length determination, the background level for the top fill channel is defined to be 20 percent amplitude. Defect interaction is not considered in length evaluation.

4. Surface defects on the outside of the pipe shall be evaluated visually and interpreted in accordance with API 1104.
5. UT interpretation of a defect in the cap notch channel as lack of fusion or undercut shall govern. This interpretation shall overrule interpretations of visual findings if the visual finding accepts the indication, but AUT rejects the indication.
6. Transverse indications other than cracks shall be considered volumetric and shall be evaluated using the criteria as described in below section.
7. Indications determined to be cracks are unacceptable.

4.13.7.6.2 Volumetric Indications

Volumetric indications shall be evaluated using the root, with +4db running gain, and volumetric, with +8db running gain mapping and A scan channels. For length determination, the background level for the root and volumetric channels shall be defined as 20 percent amplitude.

Volumetric indications are subject to the following limitations in size:

- Individual pore size or individual volumetric indication shall not exceed 3.2 mm (0.125 in.). The dimension of a single pore shall be determined from the 2-D display or from the curvature of the associated TOF channel.
- If there is buried, scattered, or cluster porosity and the signal amplitude over a given length remains over 20 percent in one or more VOL channels, then the acceptable length is the lesser of 50 mm (2 in.) in a continuous 305 mm (12 in.) length of weld or 8 percent of the weld length as if the scattered or cluster porosity was a linear indication.

- Volumetric Clusters indications are rejected if the maximum dimension exceeds 13 mm (0.5 in). The mapping channels, 2-D display and the A scan presentations shall be used to determine the dimensions. The threshold is 20 percent for the volumetric channels and 40 percent for the root channel. The background level used for length measurement is 20 percent for all channels.
- If there is buried scattered porosity and the signal amplitude over a given length remains over 20 percent in one or more VOL channels, then the acceptable aggregate length is the lesser of 50 mm (2 in.) in a continuous 305 mm (12 in.) length of weld or 8 percent of the weld length as if the scattered porosity was a linear indication. For length determination the background level for the volumetric channels shall be defined as 20 percent amplitude. If there is scattered porosity on the OD, then the aggregate length is limited to 25 mm (1 in.) in a continuous 305 mm (12 in.) length of weld. The cap notch with a 40 percent threshold relative to the 1 mm calibration notch shall be used to measure OD scattered porosity. Defect length shall be corrected for beam spread. Background level for the cap notch for length determination is defined to be 20 percent.
- Volumetric root indications, each with at least one amplitude value over 40 percent with the +4db running gain active, are rejected if the maximum dimension exceeds 6 mm (0.25 in.) or if the total aggregate length of Volumetric root indications, each with at least one amplitude value over 40 percent, exceeds 13 mm (0.5 in.) in any continuous 305 mm (12 in.) of weld. Defect length shall be corrected for beam spread. Background level for length determination is defined to be 20 percent.

4.13.7.6.3 Accumulation of Relevant Indications

Any accumulation of relevant indications shall be considered rejected when the aggregate length of all indications, each with at least one amplitude value above the threshold level, exceeds 50 mm (2 in.) in any 305 mm (12 in.) of weld or if the aggregate length exceeds 8 percent of the total weld length.

All individual indications shall be corrected for beam spread.

4.13.7.7 Repair Weld Inspection for AUT Joints

The ultrasonic system may include scanner heads and system set-up specifically configured for testing of repairs where the primary function is to confirm the complete removal of rejected flaw. As a minimum the AUT system with its normal set-up, but with gates wide enough to encompass the repair area and confirm that the AUT defects have been removed. During this special attention shall also be made to TOFD channel indications, and indications outside the normal gate settings.

Due to the wide variation in repair weld groove shapes that may limit the detection capabilities of the system, manual UT shall support the AUT on weld repairs unless the groove shape is controlled to be within given tolerances and the scanner head is configured accordingly.

4.13.7.8 Recording and Reporting

All weld inspections shall be stored at the end of each inspection scan.

AUT technicians shall maintain a System Logbook detailing the performance characteristics data and identification for instruments, transducers, and inspection setups. Register of all Calibration Blocks including unique number shall be included in System Logbook. All changes in components or procedures shall be documented in the Logbook which shall be available for Company inspection on request.

The electronic record shall be submitted to the Company on completion of the project, with the software and the key (as applicable).

Printer provides hard copies of the scanned images of calibration and the inspected welds along with all required details. It shall possess the quality of printing out the display results in a clear legible form without missing any data.

4.13.8 Liquid Penetrant Testing

Liquid Penetrant Testing of welds shall be performed according to a written procedure conforming to API 1104.

Acceptance criteria shall be in accordance with the requirements of API 1104

4.14 Add New Section: Section – SAFETY

Safety shall be of paramount importance at all times throughout the fulfilment of the requirements of this document specification. EPCm CONSTRUCTION CONTRACTOR shall observe good working practice and ensure that all activities (i.e., welding, grinding, cutting, etc.) are carried out in such a way that all risks are as low as reasonably practicable. EPCm CONSTRUCTION CONTRACTOR shall comply with all industry and country based regulatory requirements to ensure a safe working environment.

Following the contract award, EPCm CONSTRUCTION CONTRACTOR is to provide detailed policies, plans and procedures for ensuring that all safety requirements, laws, regulations, ordinances, etc., of the work being performed are strictly maintained.

5.0 REQUIREMENTS FOR CRA CLAD AND LINED PIPE GIRTH WELDING

This section specifies the girth welding requirements for CRA clad and lined pipelines. These requirements are in addition to Addendum to AGES-SP-10-006 CRA Clad and Lined Steel Pipe (Amendments Supplements to API Spec 5LD) (Document No. 30201_50150_LT_HE-ADD-SP-10-006) and ASME BPVC Sec IX as applicable.

5.1 WELDING PROCESSES

SMAW, GTAW, SAW and GMAW are pre-approved welding processes. Other welding processes may be proposed providing successful testing/qualification and COMPANY approval.

Root pass welding shall only be carried out using GTAW. Other processes shall not be used without COMPANY approval.

The welding equipment to be used in production welding shall be the same type and model of welding equipment used during welding procedure qualifications.

5.2 WELDING CONSUMABLES

Only CRA welding consumable shall be used for one-sided girth welding. The consumable selection philosophy shall be based on guidance specified in AGES-SP-10-006 CRA Clad and Lined Steel Pipe (Amendments Supplements to API Spec 5LD) (Document No. 30201_50150_LT_HE-ADD-SP-10-006).

Table 5-1 provides recommended welding consumable selection and classification for girth welding of Alloy 825 (UNS N08825) cladded pipes applicable in this PROJECT. Consumables other than those listed in Table 5-1 shall require COMPANY approval.

Table 5-1: PROJECT Consumable Selection

Clad Type	AWS/ASME Classification		
	Root & Hot Pass	Remaining Passes	
	Bare Wire	Bare Wire	Covered Electrode
UNS 08825	ERNiCrMo-3	ERNiCrMo-3	ENiCrMo-3

The ERNiCrMo-3 consumable electrode used for weld overlay shall have a maximum iron content of 1%. The iron content in the weld dilution for UNS N08825 overlay shall be limited to a maximum of 20%.

All consumables shall have clear AWS/ASME or MANUFACTURER's identification on each individual rod.

5.3 WELDING QUALIFICATION

Welding procedures shall be qualified in accordance with ASME BPVC Sect IX. For sour services application, additional requirements of Addendum to AGES-SP-07-003 (Document

No. 30201_50150_LT_HE-ADD-SP-07-003) shall be met.

Previous qualified WPS is not permitted.

All welding procedures qualification shall be carried out using the pipes intended for actual production welding.

Where repair welds may be performed in fabrication, both original weld and repair weld procedures shall be qualified separately.

In addition to essential variables stipulated in ASME BPVC Sect. IX, the following are considered as essential variables for automatic-mechanized field girth welding qualification:

- Diameter, wall thickness and grade of material.
- Bevel design and tolerances.
- Increased in internal diameter or root mismatch from PQR test coupon.
- Minimum number of passes deposited prior purging removal.
- Welding consumable brand name and wire diameter.
- Type of Line-Up clamp and time of removal.
- For welds that will be inspected by AUT, additional essential variable within Appendix C and E of DNV-ST-F101 shall apply.

In addition to the qualification test from ASME BPVC Sect IX, the following additional testing shall be performed during PQR:

1. Chemical Composition

   The sampling shall be taken at a depth of minimum 3.0mm from the ID of CRA surface (the weld overlay fusion line). Chemical composition value shall be in accordance with ASME Sec II C or AWS Specification.

2. Corrosion Testing

   Corrosion specimens shall be prepared by removing the backing carbon steel by milling or by other means. The thickness of the specimen shall not be less than 2.0 mm. No additional passivation shall be allowed.

   An intergranular corrosion test in accordance with ASTM G28 Method A shall be performed with maximum corrosion rate acceptance not exceeding 0.457 mm per year for UNS N08825.

   A pitting corrosion test shall be done in conformance with ASTM G48 Method A. No pitting shall be visible at a 20x magnification, and the weight loss shall be less than 4.0 g/m2/24 h at 22 °C for UNS N08825.

3. Macro Examinations and Hardness Testing

   Macro examination and hardness testing shall be done in accordance with Addendum

to AGES-SP-07-003 (Document No. 30201_50150_LT_HE-ADD-SP-07-003). Sample shall be taken from 6 and 12 “O” clock positions. Hardness of the CRA weld, HAZ and base metal should not exceed the limits specified in NACE MR0175/ISO 15156.

4. Micrographic Examinations

   For 5G/6G: 2 off representing min and max heat input (1 off for other positions). Microstructure of weld and HAZ to show appropriate phase distribution and absence of microphases considered deleterious to mechanical or corrosion properties. Photomicrographs to be provided for each region (typically 250X) with the WPQR.

5. Positive Material Identification (PMI)

   PMI shall be recorded during PQR execution at three locations (i.e., root, mid, and cap). Acceptance criteria shall be as per the chemistry requirement for the range of elements specified in AGES-GL-13-002.

All PQR test coupon shall be tested by NDT before any laboratory destructive testing. The NDT methods (manual or automatic) shall be the same as production weld.

5.4 WELDER AND WELDING OPERATOR QUALIFICATIONS

Welder and welding operator shall be qualified in accordance with ASME BPVC Sec IX.

5.5 PRODUCTION WELDING

The CRA surface shall have sufficient protection from any debris and contamination during cutting and beveling including direct contact with backing-shoe material.

All surfaces to be welded shall be clean and free from paint, oil, dirt, scale, oxides and other foreign material detrimental to weld integrity. The cleaned zone shall extend for a minimum of 25 mm (1 in) on either side of a weld.

Only inert purging gas conforming to AWS A5.32 shall be used. Back purge shall be maintained until at least 6 mm thickness of weld metal has been deposited or the weld joint is filled, whichever is less. Unless otherwise specified, the measured oxygen content in the purge shall be no greater than 0.05%. An oxygen analyzer shall be used for continuous monitoring of the oxygen content inside the pipe during purging.

Preheat and PWHT shall be done in accordance with applicable design code (i.e., ASME B31.8) and Section 4 of this specification (Addendum to DGS-6300-004). For sour service, additional PWHT requirement as specified in Addendum to AGES-SP-07-003 (Document No. 30201_50150_LT_HE-ADD-SP-07-003) shall apply. PWHT shall not be done without COMPANY approval.

5.6 INSPECTION

Non-Destructive Testing (NDT) procedure shall be established in accordance with this specification, ASME V and applicable design code. A written procedure for each inspection method and technique, including acceptance criteria, shall be submitted to COMPANY for

approval.

The NDT shall be 100% visual examination, 100% radiographic or ultrasonic testing, 100% PMI, and 100% liquid penetrant testing for each weld on both welding qualifications and production. In the case Automated UT is proposed, this shall comply with the requirements of DNV-ST-F101 and be demonstrated to COMPANY during procedure qualification.

100% production weld hardness shall also be carried out for sour service applications. The method and acceptance criteria shall be as per Addendum to AGES-SP-07-003 (Document No. 30201_50150_LT_HE-ADD-SP-07-003).

5.7 REPAIR

No repair shall be carried out without COMPANY approval.

Separate repair welding procedure shall be prepared for through thickness repair and partial thickness repair. The qualification testing for through thickness repair welding procedure shall be the same as the original procedure. For partial thickness repair, tensile and bend test are sufficient.

SMAW shall not be used for through thickness repair.

Unacceptable defects shall be removed by mechanical means and/or dressed down and rewelded. Thermal methods excavation such as gouging or arc-air shall not be used.

After excavation of the defect, 100% penetrant testing shall be carried out to ensure complete defect removal. The original NDT method shall be carried out after repair welding.

Repair by welding is allowed for one attempt only, re-repair (second repair) is not allowed. Repair welding shall not be performed without the use of filler metal.

APPENDIX-01 DGS-6300-004

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DESIGN GENERAL SPECIFICATION
PIPELINE WELDING
DGS 6300 004 Rev 3

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1.0 GENERAL

1.1 INTRODUCTION

This specification provides the requirements for fabrication and field welding of gas/Liquid pipelines, and related for the COMPANY facilities.

It covers the welding and NDE of pipe-to-pipe, pipe to hot bend, pipe to fitting and fitting-to-fitting welds, including multiple joint, tie-ins and attachments.

Pipeline hot taps are not included.

For pipelines in H2S sour services, additional requirements of DGS 0000-002 shall be met. For pipeline, Hot Tapping additional requirements of DGS 1300-100 shall be met.

1.2 PURPOSE

This specification supplements the codes listed in Section 2.0 of this document.

1.3 DEFINITIONS

For the purpose of this specification, the following definitions apply:

COMPANY - ADNOC GAS PROCESSING

CONCESSION REQUEST - A deviation requested by the CONTRACTOR or VENDOR, usually after receiving the contract package or purchase order. Often, it refers to an authorization to use, repair, recondition, reclaim, or release materials, components or equipment already in progress or completely manufactured but which does not meet or comply with COMPANY requirements. A CONCESSION REQUEST is subject to COMPANY approval.

CONTRACTOR - The party, which carries out all or part of the design, engineering, procurement, construction, commissioning or management of the Project. CONTRACTOR includes its approved Manufacturers, Vendors, Suppliers, and Subcontractors. Requirements of this specification shall be enforced at all levels of the CONTRACTOR'S procurement

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process. The COMPANY shall have access to all Contractor suppliers to the same extent as required of Contractor.

MANUFACTURER/VENDOR/SUPPLIER - The party(s), which manufacture(s) and/or supply(s) equipment, technical documents/drawings and services to perform the duties specified by COMPANY / CONTRACTOR.

SUBCONTRACTOR/SUBVENDOR - The party(s) which carry(s) out all or part of the design, procurement, installation and testing of the system(s) as specified by the CONTRACTOR/VENDOR and/or COMPANY.

DRAWINGS - Drawings provided by the CONTRACTOR/VENDOR.

PROJECT - (To be defined).

PROJECT MANAGEMENT TEAM (PMT) - The COMPANY authorized party responsible for overall day-to-day execution of the PROJECT. The PMT is to serve as the liaison between the COMPANY and the CONTRACTOR(s) on the PROJECT.

SHALL - Indicates a mandatory requirement.

SHOULD - Indicates a strong recommendation to comply with the requirements of this document

2.0 CODES AND STANDARDS

The following codes and standards, to the extent specified herein, form a part of this specification. When an edition date is not indicated for a code or standard, the latest edition in force at the time of CONTRACTOR'S/VENDOR'S proposal submittal shall apply.

AMERICAN WELDING SOCIETY (AWS)	
ANSI/AWS A5.01	Filler Metal Procurement Guidelines
AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)	
ASTM 370	Standard Test methods and definitions for Mechanical testing of Steel products.

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ASTM G39 ASTM E92	Preparation and Use of Bent-Beam Stress-Corrosion Test Specimens. Standard Test Methods for Vickers Hardness of Metallic Materials.
AMERICAN PETROLEUM INSTITUTE (API)	
API 5L API 5LD	Specification for Line Pipe CRA Clad or Lined steel Pipe
API 1104	Welding of Pipelines and Related Facilities
AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)	
ASME B31.8 ASME B31.4 ASME B16.49	Gas Transmission and Distribution Piping Systems Pipeline Transportation Systems for Liquid Hydrocarbon and other liquids. Factory Made Rough Steel, Butt Welded Induction bends for Transportation and Distribution systems.
NATIONAL ASSOCIATION OF CORROSION ENGINEERS (NACE)	
NACE MR0175/ ISO15156-Parts 1/2/3	Petroleum & Natural Gas Industries Materials for use in H2S containing environment in Oil & Gas production.
NACE TM0177	Laboratory Testing of Metals for Resistance to Sulfide Stress Cracking and Stress Corrosion Cracking in H2S Environments
AMERICAN SOCIETY OF NONDESTRUCTIVE TESTING (ASNT)	
RP SNT-TC-1A	Personal Qualification and Certification in Non-Destructive Testing
BRITISH STANDARD	
BS 7448, Part 1, Part 2 BS EN ISO 15653	Fracture mechanics toughness tests – welds. Metallic Materials – Method of tests for the determination of quasistatic toughness of welds.
EUROPEAN STANDARD	
EN 10204	Metallic Products – Types of inspection documents
INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)	
ISO 9001	Quality Management Requirements
ISO 9004	Quality management -- Quality of an organization -- Guidance to achieve sustained success

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3.0 REFERENCE DOCUMENTS

The following reference documents, to the extent specified herein, form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of CONTRACTOR'S proposal submittal shall apply.

DESIGN GENERAL SPECIFICATIONS

DGS 0000 002	Materials for use in H2S Containing Environment.
DGS 6300 005	Requirements for radiographic Inspection of Pipeline welds.
DGS 9510-003	High Frequency Electric Welded (HFW) Line Pipes.
DGS 9530-001	Factory Hot Bends.
DGS 9530-002	Pipeline Pigging Tees.
DGS 9540-001	Pipeline Scrapper Launcher and Receivers.
AGES-SP-10-001	Specification for Linepipe
AGES-SP-10-002	Specification for Subsea Pipeline systems
AGES-SP-10-003	Onshore Pipelines Design and Construction Specification
AGES-SP-09-003	Piping & Pipeline Valves Specification
AGES-SP-07-002	External Pipeline Coatings Specification

4.0 DOCUMENT PRECEDENCE

It shall be the CONTRACTOR'S responsibility to be, or to become knowledgeable of all of the requirements of the referenced Design General Specifications, Procedures, Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein. Resolution and/or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design or manufacture.

In case of conflict, the order of precedence shall be:

- Description of Project Scope and statement of requirements
- Project Specifications and Drawings

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- This Specification
- Industry Codes and Standards

5.0 SPECIFICATION DEVIATION/CONCESSION CONTROL

Deviations from this specification are only acceptable where the CONTRACTOR has listed in his quotation the requirements he cannot, nor does not wish to comply with and the COMPANY has accepted in writing the deviations before the order is placed.

In the absence of a list of deviations, it will be assumed that the CONTRACTOR complies fully with this specification.

Any technical deviations to the Purchase Order and its attachments including, but not limited to, the Data Sheets and Narrative Specifications shall be sought by the CONTRACTOR only through CONCESSION REQUEST format. CONCESSION REQUESTS require COMPANY'S review/approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

6.0 QUALITY ASSURANCE/QUALITY CONTROL

A copy of CONTRACTOR'S QA/QC management program shall be submitted to COMPANY for review and concurrence. If CONTRACTOR'S QA/QC program is ISO 9001 certified, then only a copy of the ISO 9001 certificate is required. In addition, if CONTRACTOR'S facility is ISO 9001 certified, QA audit requirements will be waived in favor of ISO 9001 registrar audits, unless COMPANY'S trend analysis program indicates areas of concern. Unless otherwise agreed with the COMPANY, the CONTRACTOR'S Quality System shall fully satisfy all the elements of ISO 9001,"Quality Management Systems - Requirements" and ISO 9004 "Quality management -- Quality of an organization -- Guidance to achieve sustained success".

Notwithstanding ISO 9001 accreditation, CONTRACTOR shall submit the necessary Quality Plans, QC Procedures, Method of Statements, NDT and Welding Procedures as required by COMPANY and specific to the project work scope prior to the commencement of work.

CONTRACTOR shall identify in documents to its manufacturers, suppliers, vendors and subcontractors all applicable QA/QC requirements imposed by COMPANY, and shall ensure

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compliance. On request, CONTRACTOR shall provide objective evidence of its QA/QC surveillance for all levels of its activity.

COMPANY/CONTRACTOR reserves the right to inspect materials and workmanship at all stages of

Manufacturing and to witness any or all tests. The vendor, 30 days upon award but prior to the pre-inspection meeting shall provide the CONTRACTOR with a copy of its manufacturing Inspection and Test Plan and with copies of all related/referenced procedures for review and Acceptance. The inspection and test plan will also be reviewed for inclusion of any mandatory COMPANY/CONTRACTOR/EPC CONTRACTOR Witness or hold points. The laboratory which conducts chemical analysis and the various mechanical tests required by this specification shall also have ISO 9001 accreditation and preferably industry specific Accreditation

7.0 DESIGN CONSIDERATIONS

The following climatic conditions shall govern the design of the equipment:

Temperature:

Maximum in shade	:	58°C
Maximum in the sun	:	85°C
Minimum ambient	:	4°C

Relative Humidity:

Maximum (at 43°C)	:	95 percent
Average (at 28°C)	:	60 percent
Design	:	100 percent

Rainfall is infrequent but may occur in winds up to 40 m/sec.

Sandstorms are frequent and dust adheres to all exposed surfaces.

The Pipeline and associated facilities shall meet the requirements of NACE MR0175/ ISO15156-1 when sour service condition is specified.

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8.0 WELDING EQUIPMENT, TOOLS AND SUPPLIES

All welding machines, line up clamps, beveling machines, cutting torches, and other equipment, tools, and supplies used in connection with the welding work shall be kept in good mechanical condition. Sufficient meters, portable or permanent, shall be provided to calibrate machines and to regulate voltage and amperages as may be needed. Ground clamps shall be dependable and not arc the pipe. The clamps shall be located close to the point of welding.

The welding equipment used in production welding shall be the same type and model of welding equipment used during welding procedure qualifications.

A detailed procedure for operation of Automatic Welding (not SAW) equipment outlining set-up and sequence of operation shall be submitted for COMPANY approval. It shall include:

- Input welding parameter detail and max/min programmed tolerances
- Method statement for the control of the welding parameter cartridges at all weld stations
- Max/min welding operator adjustment control limits.
- Procedure for adjusting welding parameters at the welding console, including report sheet.

Any equipment which, in the opinion of COMPANY is not adequate for the welding to be done, shall be replaced by CONTRACTOR.

Qualification of welding procedure for automatic welding shall be based on the actual field welding process to be adopted by the contractor in respect of the following:

a) Number of welders per joint

b) Number of machines per joint

c) Sequence of welding

Any change to the above parameters during production welding shall require re-qualification of the welding procedure qualifications. If more than one machine per joint is to be used for WPS qualification it must simulate the actual site sequence.

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9.0 PIPELINE MATERIALS

The pipe, hot bend and fitting materials shall be as specified in the Contract Specifications.

Line pipe residual magnetism may be 25 gauss maximum. CONTRACTOR shall ensure residual magnetism does not impede welding.

10.0 ELECTRODES, FILLER WIRES AND FLUXES

All welding consumables shall be specified in COMPANY approved welding procedure specifications (WPS), including the Manufacturer's name and brand.

All welding consumables shall be stored and handled in accordance with the Manufacture's recommendations, as a minimum. CONTRACTOR storage, issue and handling procedures for welding consumables shall be submitted to COMPANY for review and approval. Consumables which become wet, contaminated or damaged shall be removed from the Project Site and destroyed.

Each tin or packet of electrodes, or reel of filler wire and bag of flux shall be clearly marked with the respective lot number. Manufacturer's Certification shall be supplied to EN10204, 3.1, for each lot of welding consumables according to ANSI/AWS A5.01;

- For SMAW – Lot classification C4, Schedule J.
- For GMAW/SAW wire – Lot classification S2 from one heat, Schedule J.
- For SAW flux – Lot classification F2, Schedule J.

The Nickel content of welding filler metals and fluxes shall not exceed 1% for sour service applications

The Molybdenum content of filler metals and fluxes shall not exceed 0.5% and shall not be deliberately added.

Nickel and Molybdenum content shall be included in Manufacturer's Certification.

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All welding consumables used during production shall be the same manufacturer, brand and AWS type as those used and approved during welding procedure qualifications. A full set of mechanical tests shall be conducted in accordance with this specification.

Sufficient quantities of each lot of welding consumables should be purchased by CONTRACTOR to complete the work.

CONTRACTOR shall maintain a system of traceability whereby the batch numbers of the welding consumables are related to the weld numbers with that particular joint.

11.0 SHIELDING GASES

When specified in the WPS the shielding gas or gas mixture to be used and gas quality shall be qualified in the welding procedure qualification record.

Shielding Gases Purity and Dew point shall be as per SFA 5.32 (specification for Shielding Gases).

Where mixed gases are to be used the specification composition and purity are to be approved by COMPANY.

Each container of gas shall be identified with the gas type, mixture and purity. Unmarked containers and/or contaminated gases shall be removed from the work site. The shielding gas manufacturer shall issue a test certificate qualified purity levels for each batch / lot.

12.0 QUALIFICATION OF WELDING PROCEDURES

12.1 GENERAL

CONTRACTOR shall submit detailed welding procedure specifications for approval by COMPANY for each pipe grade, size, and wall thickness - see 12.4(b), prior to commencement of welding procedure qualification tests.

Welding procedures are required for all production welding including butt welds, fillet welds, and all associated repairs.

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Separate welding procedures shall be submitted and qualified for the following:

- Pipeline welding.
- Tie-in welding.
- Welding of fittings.
- Welding of weldolets.
- Repair welding.
- Fillet welds.
- Attachment of saddles/split sleeves.
- CP attachments.
- Cap, partial penetration and full penetration repair procedures shall be qualified. See Attachment 3, Figure 3 for location of partial penetration weld repairs.
- The following welding processes are pre-approved:
- Gas Tungsten Arc Welding(GTAW) – Manual
- Shielded Metal Arc Welding (SMAW) - Manual
- Submerged Arc Welding (SAW) - Automatic
- Gas Metal Arc Welding (GMAW) – Automatic, subject to CONTRACTOR having a successful past history using the proposed automatic GMAW process and/or system and the following:
  - A pre-weld qualification test is carried out on contract pipe to demonstrate that service requirements defined in this specification are fully achieved.
  - Automatic parameter monitoring equipment for measuring amperage and voltage at the welding heads to verify programmed parameters are being achieved.
  - All welding operators have been qualified on the machine equipment and have worked on production welding on prior Projects using the process.
  - 100% UT testing, in addition to RT, is carried out on the pre-weld qualification test welds.
  - A detailed UT procedure is prepared for the pre-qualification welds, to include a test weld with known defects, which will be used for subsequent weld procedure qualification and pipeline welding.
  - Control of welding parameters is clearly defined – Clause 8.0.
  - Bevel tolerances are specified.

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- Preheat trials are included.
- Welding consumables are clearly specified and controlled.
- Information is supplied on previous Project repair rates/week and X-radiographic and UT NDE results. This documentary evidence shall be subject to COMPANY review.

Electrode polarity shall be considered an essential variable for qualification of SMAW procedures intended for vertical-down root pass welding.

In each case where an automatic welding process or system is approved for use, a manual welding procedure shall be qualified as a back-up.

Roll welding shall only be acceptable when using a fully automatic welding process.

Oxyacetylene welding and the flash butt welding process shall not be permitted.

The maximum time lapse between the root and hot pass (second pass) shall be recorded on the welding procedure specification and shall be qualified during weld procedure qualification. The time shall be kept to a minimum

The Weld direction shall be downhill except for any of the following:

1. Repairs
2. Tie-in welds
3. Joints where an external line-up clamp is used.

For the above welds the root bead shall be deposited in the uphill progression.

No single pass welds, or internal root sealing welds shall be allowed, except for an automatic root pass applied internally that is an integral part of the welding procedure.

Preheat shall be specified on the Welding Procedure Specification when it is required and shall be qualified during weld procedure qualification testing. The following methods of preheat are acceptable:

- Electric induction

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- Electric resistance
- Oxy-fuel gas

The use of oxyacetylene gas for preheating is not permitted.

When oxy-fuel gas is used for preheating, full encirclement rings are preferred. If hand-held torches are approved by COMPANY, a minimum of two oxy-fuel gas torches shall be used on pipe diameters greater than 6 inch NPS, when preheating manually. Oxygen cutting torches are not permitted. For thickness >18mm and/or diameter >36" Induction / electrical heating is mandatory.

All tie-in welds, fillet welds, pipe to hot bends, and fitting-to-pipe welds shall be preheated to 50°C minimum above that required for similar pipe-to-pipe butt welds. For repairs, see Clause 14.5. The actual preheat shall be qualified during weld procedure qualification. The interpass temperature shall not exceed 250°C.

If post weld heat treatment (PWHT) is required, it shall be specified on the WPS and shall be qualified during weld procedure qualification. A separate PWHT procedure shall be submitted to Company for approval, and shall include thermocouple placement and PWHT cycle. In case of Q&T, TMCP material, the steel manufacturer shall specify the construction PWHT temperature range in the Manufacturing Procedure Specification taking into consideration the PWHT temperature range specified in the construction code. For pipeline which requires PWHT, simulated coupon from raw material shall be subjected to one cycle of PWHT. These simulated coupons shall be tested and shall meet all relevant material requirements.

Acceptable PWHT methods are:

- Electrical induction
- Electrical resistance

Where pipes or fittings of unequal internal wall thickness are to be welded, the end preparations shown in ASME B31.8, Appendix I shall be used. For unequal external wall thicknesses, the weld taper shall form part of the weld procedure qualification.

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CONTRACTOR shall be responsible for developing and obtaining COMPANY approval of all welding processes, equipment and procedures.

CONTRACTOR shall notify COMPANY of the date and location of the Welding Procedure Qualification Test welding at least five (5) working days before the start of any test. COMPANY reserves the right to witness all Welding Procedure Qualification and Welder Qualification welding and testing.

All welding procedures qualification welding ,Welder performance qualification welding , associated NDE and mechanical testing shall be witnessed , test reports and test results reviewed for conformance to approved Project requirements by an independent Inspector from Third party Inspection Agency approved by COMPANY and appointed by CONTRACTOR. The above approved Inspector shall also review all the duly completed PQR /WPS documents of CONTRACTOR and endorse the compliance to COMPANY approved Project requirements prior to submittal for COMPANY approval."

12.2 RECORDS

Each revision of a welding procedure specification shall be given a new number in sequential order and revisions marked in order to identify changes.

12.3 WELDING PROCEDURE SPECIFICATION

In addition to the requirements of Section 5.3 of API 1104, the following minimum details shall be included on the Welding Procedure Specification:

1. Project Identifications.
2. Consumables trade names, type and lot number.
3. Weld preparation and fit up tolerances.
4. Preheat temperature and method.
5. Time lapse between end of root bead and start of hot pass (second pass).
6. Interpass temperature (maximum).
7. Nozzle diameter for automatic welding.
8. Oscillation width and frequency for automatic welding.

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9. Electrode stick-out length.

12.4 WELD PROCEDURE QUALIFICATION

All welding procedures shall be qualified on project material.

Pre-qualified welding procedures are not permitted.

Welding procedures shall be qualified under simulated conditions similar to the planned installation, using the same welding, alignment and preheat equipment that will be used in production. Sufficient test welds shall be made to include repair procedure qualification and all required mechanical testing.

In addition to those in Section 5.4 of API 1104 the following essential variables shall be added:

a) A change from seamless to welded pipe and vice-versa.
b) Any change from one material grade to another
c) A change in grade of, pipe, diameter, or wall thickness grouping (as identified in API 1104 paragraph 6.2.2 Items D and E) or source of supply or manufacturer. For automatic GMAW, each change in wall thickness.
d) A change in the process of base material manufacture or in the heat treatment condition (normalized, quenched and tempered.)
e) Any change in size, manufacturer, trade name or AWS classification of electrode or filler wire.
f) Any change in type of current or polarity.
g) Any change in heat input beyond +/-10 percent of the range established during the welding procedure qualification.
h) Any decrease in preheat from that recorded during welding procedure qualification. The preheat may be increased by up to 50° C without re-qualification, subject to maximum inter pass temperature limitations.
i) Any change in the minimum number of weld layers.
j) Any change in the number of root pass welders.
k) Any increase in the as qualified time delay between the root and hot pass (second pass).
l) Any change in the qualified joint design tolerances.

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m) Any change in shielding gas and flow rate.
n) Any change in shielding flux.
o) Any change in max interpass temperature.
p) A change in Post Weld Heat Treatment.
q) Any change in order of cap bead deposition.
r) Any change in welding equipment for automatic process.

For pipe to pipe qualification, full pipe lengths shall be used, including the use of a proprietary internal line-up clamp. Lowering off shall be included in the qualification.

Pipe supplier to provide histogram of Carbon equivalents for all pipes supplied. The Carbon equivalent of pipes used for welding shall not increase by 0.03 above the Carbon equivalent of Line pipe/ Forging/ fitting used procedure qualification testing.

For the qualification of hot bends and fittings welds, fittings or test rings can be used.

Cap, partial penetration and full penetration repair procedures shall be qualified.

For fillet welds, the minimum and maximum fillet weld sizes shall be qualified. Basic low hydrogen weld consumables shall be used with a minimum of two weld passes.

For mechanized welding equipment, other than SAW, a minimum of three consecutive complete test joints are to be made. Automatic monitoring equipment shall be available for measuring amperage and voltage at the welding heads to verify programmed parameters are being achieved.

A 1.6mm root misalignment, if necessary using a 1.6mm shim, shall be induced in one of the three welds.

All three joints shall pass the required NDE. One of the welds shall be used for repair qualifications. Cleaning between weld passes shall not be required if demonstrated satisfactorily during weld procedure qualification.

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12.5 WELD PROCEDURE QUALIFICATION TESTING

a) All procedure qualification test welds shall be allowed to cool for a minimum of 24 hours in simulated site conditions. The welds shall be visually inspected and then fully X-radiographed prior to destructive tests. If X-radiography indicates the presence of flaws exceeding the levels permitted in this Specification, the reason for these flaws shall be investigated and explained to the satisfaction of COMPANY before re-welding a test piece to the same (or any other) welding procedure. GMAW automatic welding shall be additionally fully inspected in accordance with an ultrasonic testing procedure, approved by COMPANY.

b) The qualification welds shall be destructively tested in accordance with API 1104, except as amended by this Specification.

c) Tensile test yield and tensile strength shall not be less than the Min. specified tensile strength requirements of the line pipe material specification used for the Procedure Qualification Tests. Maximum tensile strength shall not exceed the minimum specified tensile strength of the pipe plus 30 percent.

d) Three transverse macroscopic sections shall be taken from each weld close to the 12 o'clock, 3 o'clock and 6 o'clock positions. They shall be prepared to a diamond paste finish and etched to show the weld metal, heat-affected zone and base metal micro-structures.

   Photomicrographs at X200 showing cap and root bead weld metal and heat affected zone micro-structures shall be submitted with PQR's for all weld procedures, including repairs. Untampered martensite is unacceptable.

   Vickers hardness testing shall be carried out in accordance with Attachment 1, Figure 1. The maximum hardness shall not exceed 248 HV10 for sour service applications and for non sour, hardness shall not exceed 300 HV10.

   Photomacrographs at X5 showing the hardness indentation locations shall be submitted with the Welding Procedure Qualification Record.

e) Charpy Impact test sets in accordance with ASTM A370 shall be taken from the 12 o'clock, 3 o'clock and 6 o'clock positions as follows:

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- Centre of Weld
- Fusion Line
- + 2 mm from fusion line in the parent metal

Each test set shall include 3 specimens.

For rolled welds, the 12 o'clock position shall be the finish point of the external weld.

For material over 20 mm nominal thickness, Charpy impact tests shall be taken from the cap and root region.

The Charpy V-notch impact test temperature shall be as per Table (2)

Attachment 2, Figure 2, of this Specification shall define notch location of impact test specimens.

Minimum Charpy impact energy requirements shall be in accordance with Table 1 of this Specification.

f) Fillet welds shall be qualified by a transverse weld macro section. The section shall be etched and shall show complete fusion with the base metal. The maximum hardness shall not exceed 248 HV10 for sour service applications and 300 HV10 for all other applications. Attachment 1, Figure 1, shall establish hardness measurements.

g) Pipe grades API 5LX52 and higher for sour service conditions will require sulphide stress corrosion cracking (SSCC) tests shall be carried out on all weld and repair procedures. The SSCC procedure shall be in accordance with NACE TM0177, Method B, using solution B, and transverse weld four point bend test specimens as in ASTM G 39. Applied deflection shall correspond to the specified minimum yield strength of the base material. Tests shall be taken from the 6 o'clock position to include the weld metal and HAZ from the weld root, mid thickness and cap. Total 12 specimens. There shall be no cracking after 720 hours exposure.

h) Crack Tip Opening displacement:

I. Crack tip opening displacement (CTOD) tests shall be conducted at 0 degree C or design temperature ,whichever is lower ,for all welding procedure qualification tests

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(except Fillet welds) of Gas pipelines with pipes having thickness of 12.5 mm (½ inch) and above to ensure integrity of girth welds.

II. As the fracture toughness shall be regarded as size sensitive , qualified for the thickness tested, the CTOD values shall include thickness tested. The tests shall be conducted in accordance to BS 7448 Part 1 and BS EN ISO 15653:2010 using SENB (Single Edge Notched Bend) geometry test specimens.

III. For each welding procedure , both the weld metal and Heat affected Zone (HAZ) at 12 "O" clock, 3 or 9 "O" clock and 6 "O" clock positions shall be tested as per API 1104 , Appendix A, Section A.3.2.3.3. Each test (of weld or HAZ) shall consist of at least three valid specimens and the tests performed at 0 degree C or design temperature, whichever is lower. The three specimens shall consist of one each from the nominal 12,3 or 9, and 6 'O' clock positions on the test weld and should be permanently marked to identify the original position .

The test report shall include all items specified in Section 13 of BS EN ISO 15653:2010.

IV. The acceptance criteria for the fracture toughness of all six specimens in terms of the CTOD values shall be that the value for all six specimens, as a minimum, shall exceed 0.005" (0.15mm) at the test temperature . However CTOD testing results acceptance criteria shall be reconfirmed with COMPANY for specific Pipeline on the basis of the pipeline design criteria prior to Welding Procedure Qualification activities.

V. The test reports shall include the load –displacement record and clear photographs of both fractured faces , and shall be submitted with PQR documents for COMPANY review.

VI. The test reports, including analysis of outcome of the

CTOD test results specific to the Pipeline, and the determination of acceptable imperfection size computed using guidelines provided in Section A.5 of Appendix A of API 1104 and or other relevant International standards/codes and a graphical plot /curves of allowable imperfection size should be provided for Company information.

The CTOD testing criteria shall be agreed with COMPANY and the results shall be submitted for information.

i) CONTRACTOR shall review, witness and submit for approval all non-destructive and destructive testing procedures. All mechanical testing shall be carried out by a certified testing facility subject to the approval of COMPANY.

j) All procedure Qualification welding and testing shall be witnessed and certified by an Independent inspection authority approved by the COMPANY.

k) A register of all welding procedures to be used in production, shall be maintained by CONTRACTOR prior to the start of any production welding, and shall be subject to COMPANY approval.

l) For partial penetration weld repair qualification welds, the following mechanical tests shall be taken:

- Tensile tests – 2 specimens
- Side bend tests – 2 specimens
- Macro and hardness survey – one specimen
- Charpy test – to specification
- SSCC test – 3 specimens from base of repair

For cap repairs:

- Side bend tests – 2 specimens
- Macro and hardness survey – one specimen

TABLE1

STEEL GRADE	SPECIMEN SIZE (MM)	CHARPY -V NOTCH ENERGY REQUIREMENT (JOULES)	
		average 3 specimens	single any 1 specimen
	10 X 10	27	20
Grade B	10 X 6.7	21	16
	10 X 5.0	18	13
X42 to X60	10 X 10	41	31
	10 X 6.7	32	24
Fittings/Plate	10 X 10	To Project Fittings	To Project Fittings
	10 X 6.7	Specification	Specification
X 65	10 X 10 10 X 6.7	45 35	34 27
X 70	10 X 10 10 X 6.7	48 37	36 28

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TABLE 2 - CHARPY V-NOTCH TEST TEMPERATURE

NOMINAL WALL THICKNESS (T) (MM)	TEST TEMPERATURE (°C)
T ≤ 16	T
25 ≥ t > 16	T - 10
T > 25	T - 20

NOTE: T = minimum design temperature.

13.0 WELDER QUALIFICATION

13.1 QUALIFICATIONS OF WELDERS

All welders and welding operators employed by CONTRACTOR shall be required to pass a welding test for the type and methods of welding he will perform, including attachment welds. No welder shall work on any phase of the pipeline unless he has been tested and approved by COMPANY. Testing shall be in accordance with API 1104 Section 3. For pipe diameters over 12 ¾" a welder can be qualified on half the pipe circumference, this shall include the 12 and 6 o'clock positions. Automatic Welding operators must have previous experience / qualification on proposed equipment prior to testing. All automatic-welding operators will require requalification before production on all the appropriate pipe sizes / thicknesses.

Testing of prospective welders shall be at CONTRACTOR'S expense. CONTRACTOR shall furnish all welding materials and equipment, which shall be the same as those which will be used for production welding. CONTRACTOR shall supply prepared test nipples/test pieces from project materials.

Welder tests shall simulate production conditions, with the distance between the bottom of the pipe and the ground being between 40-60cms.

The test welds shall be examined and tested by CONTRACTOR, and witnessed and approved by COMPANY.

The quality of the test welds shall be determined by radiographic testing, using the approved RT procedure, and in addition, for GMAW welding, the approved UT procedure, and shall

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meet the acceptance limits of Clause 15.5. Welds passing NDT shall be finally qualified by destructive testing to API 1104.

Each welder/operator shall be given a unique number which shall be marked in paint on all welds produced by him. Paint quality shall approved by COMPANY.

A list of qualified welders shall be kept up to date and available for COMPANY inspection. This shall include a welder's competence certificate which references the corresponding WPS number, the essential variables, validity date and the test results for each welder or welding operator for each test.

Should a welder or welding operator leave the job, his number shall not be duplicated.

Any welder found responsible for an unacceptable level of defects/repairs, shall, at COMPANY request, be removed from production. At the discretion of COMPANY, the welder in question may be retested after an agreed upon period of retraining.

14.0 PRODUCTION WELDING

All welding shall be performed in accordance with API 1104 and this specification.

14.1 GENERAL REQUIREMENTS

Welding requirements shall be as specified in the API 1104 and this specification, including:

- Pipe end preparation, bevels and cleaning
- Alignment
- Line-up clamps and pipe supports
- Working clearances
- Weather conditions
- Earth clamps
- Tack welds
- Inter-run cleaning
- Partially completed welds
- Stray arcs

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- Pre and post weld heat treatment
- Maximum interpass temperature

On pipe which is to be cut back or sectioned, a zone extending 100mm from the proposed position of the new field bevel shall be examined visually and by ultrasonic testing to the acceptance requirements of the appropriate pipe material specification. In addition, the finished new bevel shall be magnetic particle tested. This does not apply for initial pipe end beveling for automatic GMAW.

During welding production, each automatic welding head (not SAW) shall have its welding parameters checked on a random basis by the approved metering equipment – see Clause 12.4. All results are subject to COMPANY approval.

All field bevels on line pipe, and end preparation for unequal wall thickness, shall be made with a beveling machine. Hand beveling shall not be permitted.

14.2 PIPE CLEANING

The following requirements shall be met by CONTRACTOR:

- Each joint of pipe shall be swabbed along its full length, immediately prior to alignment with the pipeline.
- Swabbing shall remove all loose dirt, mill scale, and other foreign substances inside the pipe.
- The swab shall incorporate a steel disc, minimum 0.25 inch (6.4mm) thick, with a diameter 0.4 inch (10mm) less than the nominal inside pipe diameter (ID).
- The event that the steel disc will not pass through a joint of pipe with the plane of the disc at right angles to the pipe wall, that joint shall be rejected and shall not be installed in the pipeline. Each rejected joint of pipe shall be clearly marked as a reject and removed from the jobsite.

14.3 ALIGNMENT OF PIPE

Details of alignment clamps shall be submitted to COMPANY for approval. All joints shall be aligned with internal line-up clamps, except for tie-in's or under exceptional conditions as

approved by COMPANY. In situations where an internal clamp is impractical, an external line-up clamp may be used.

Longitudinal seams in adjoining pipe lengths shall be staggered by not less than 20 degrees and shall be located in the top 120 degrees of the circumference of the pipe. For all bends, the longitudinal seam shall be in the plane at right angles to any bend.

Maximum misalignment or offset shall not exceed 1.6 mm prior to welding the root bead.

Force shall not be used to correct misalignment other than the normal force associated with the use of conventional line-up clamps.

14.4 WEATHER LIMITATIONS

The welding procedures approved by COMPANY shall be strictly adhered to at all times with regard to welding parameters and environmental conditions. Welding may not be performed when, in the judgment of COMPANY, the quality of the completed weld would be impaired by the prevailing weather conditions. These conditions include, but are not limited to: airborne moisture, sand or dirt particles and high winds. Windshields and enclosed welding stations may be used when practical.

14.5 PRE-HEATING AND INTERPASS TEMPERATURE AND POST-WELD HEAT TREATMENT

Preheating and interpass temperature requirements shall take into account weldability results that are available from the line pipe supplier, and shall be qualified in welding procedure qualification testing. For automatic welding (not SAW) preheat trials are required to ensure specified hardness requirements are obtained.

Unless otherwise agreed by COMPANY, the preheat for repair welds shall not be less than that qualified for the original weld, but in no case shall be less than 100°C, and shall be applied uniformly around the full circumference of the welds

Any requirements for post-weld heat treatment shall be established during welding procedure qualification tests. In such cases, the specific requirements for heat treatment shall be included in the welding procedure specification.

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14.6 TACK WELDS

Tack welds when approved by COMPANY shall be a minimum of 50 mm long, and shall be welded with the same Welding Procedure Specification used for the root pass and by a qualified welder.

All tack welds shall be welded vertical-up.

Preheat of tack welds shall be the same as the approved welding procedure specification for that pipe or fitting grade, size and thickness.

Tack welds incorporated into the final weld shall be ground to a feather edge at each end to ensure adequate fusion on the root pass.

14.7 LINE-UP CLAMPS

Internal line-up clamps shall not be removed before the completion of the first pass.

The pipes on both sides of the weld joint shall not be raised, lowered, or otherwise moved, until both the root pass and the second pass (hot pass) have been completed, unless CONTRACTOR can demonstrate to the satisfaction of COMPANY that the root pass alone is adequate to resist deformation and cracking. This shall include a stress calculation.

External line-up clamps may be removed when the root pass is at least 50 percent complete and equally spaced around the pipe circumference, subject to the pipe remaining adequately supported. For root bead welding, refer to Clause 12.1.

14.8 INTER-RUN CLEANING

Each weld pass shall be cleaned to remove all welding slag and foreign matter prior to welding successive passes. Cleaning methods shall be in accordance with the approved Welding Procedure Specification. For automatic welding, the cleaning inter-run cleaning requirement will be reviewed dependent on NDT results.

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14.9 ARC STRIKES

Arc strikes shall not be permitted outside the weld bevel.

14.10 MULTIPLE JOINT WELDING

Multiple joint welding may be performed either manually or by automatic processes, providing the process is in accordance with a qualified procedure approved by COMPANY.

14.11 MARKING WELDS

All welds shall be given a unique number in sequence. The sequence shall be recorded on identification weld reports and/or field drawings. The weld numbering sequence shall be submitted to COMPANY for approval.

Prior to radiographic inspection, CONTRACTOR shall paint the weld number on the pipe adjacent to the weld at a point where it shall remain visible after field coating. The numbers shall be applied with a spray can or brush using water-proof paint of a contrasting color to the pipe coating. Punch or steel stamping shall not be permitted. The weld numbers shall be a minimum of 100 mm high and legible.

14.12 PROXIMITY OF WELDS

Adjacent circumferential welds in pipelines shall be separated by a minimum of 1.5 meter Pipeline.

The distance can be reduced to 500 mm in the above ground pipe at valve stations and pipe on the plant side of anchor blocks or expansion loops at end stations.

Branches, fittings, and attachments shall be separated away from other welds on the pipeline. The minimum separation shall be the greater of:

- Diameter of Branch/Fitting
- Six (6) times the wall thickness of the thickest component
- 150 mm

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14.13 END PROTECTION

Pipe ends shall be adequately protected during transportation, handling and fabrication and installation.

Bevel protectors, where provided, shall remain in place until fabrication commences.

On completion of stage fabrication or end of work shift or end of the day, secure end caps or plugs shall be placed over open ends. Welding to pipeline material is not permitted. End caps and plugs shall be approved by COMPANY.

15.0 STANDARDS OF ACCEPTABILITY

The standards of acceptability for all welding shall be in accordance with the requirements of API Standard 1104, Section 6, and this Specification.

15.1 INSPECTION AND TESTING – GENERAL REQUIREMENTS

CONTRACTOR shall provide all weld inspection.

Each weld shall be cleaned by a power brush to remove all slag, splatter, etc., before inspection.

All welds shall be inspected 100 percent by visual examination.

All butt welds shall be inspected 100 percent by radiographic inspection.

See Clause 15.3 for the Ultrasonic Testing (UT).

Non-destructive Testing (NDT) procedures shall be submitted for COMPANY approval and shall be qualified individually on the welding procedure qualification tests, or when agreed to by COMPANY, on a production weld.

Where geometry of the welds does not allow radiographic testing, ultrasonic testing or other approved non-destructive testing methods may be used. Such instances shall be approved by COMPANY.

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All NDT operators/technicians shall be qualified to ASNT Level II or equivalent and shall have a minimum of 2 years experience on pipelines working in the NDT grade qualified. All operators certification will be subject to approval by the COMPANY.

For UT at least one level III qualified Inspector shall be available for overall supervision.

The final non-destructive testing of welds for acceptance purposes shall be carried out after completion of any PWHT.

All the results of any radiography examination carried out before PWHT by the VENDOR shall be submitted as well for the review of the CONTRACTOR. Please note this does not remove the requirement for final acceptance NDT examinations required after PWHT.

15.2 RADIOGRAPHIC INSPECTION

CONTRACTOR shall supply all labor and materials necessary to perform radiographic inspection. Any weld which in the COMPANY's opinion is not satisfactory as indicated by radiographic inspection shall be repaired or cut-out and re-welded at CONTRACTOR'S expense.

Radiographic inspection shall be in accordance with DGS 6300 005.

15.3 ULTRASONIC INSPECTION

Ultrasonic examination is required as follows:

a) After re-bevelling due to pipe cutting, pipe damage and/or cut outs (not initial pipe end re-bevelling for automatic GMAW welding). A 100mm wide band extending from the pipe end.
b) A zone 100mm wide around planned cut-outs for nozzles, branches etc.
c) To verify minimum wall thickness after grinding.
d) All restrained tie-in welds
e) Initial testing of automatic welds.

At the commencement of welding with an approved automatic process (other than SAW) all welds shall be tested 100% by ultrasonic examination until COMPANY is satisfied that the

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process is under complete control (minimum of 50 welds). The level of UT examination may then be reduced with COMPANY's approval, when a satisfactory level of confidence in the automatic process and quality of the welding has been reached.

A detailed UT procedure is required which includes a test block with known defects for calibration purposes – refer to Clause 12.1

f) Where radiographic inspection does not clearly show compliance with this specification.

Excluding (iv) and (v), acceptance shall be in accordance with the relevant pipe material specification.

Ultrasonic operators shall be certified in accordance with ASNT Level II. COMPANY reserves the right to test proposed UT operators prior to acceptance using the approved procedures on a test weld.

Ultrasonic inspection procedures shall be submitted to COMPANY for approval, prior to use.

For automatic UT and conventional UT inspection, a detailed procedure shall be prepared. Prequalification on a test girth weld made with known defects of required sizes as per sensitivity requirement using the same pipe grade, OD and thicknesses of the line pipe of the main line project.

For Automatic UT (AUT):

AUT is acceptable for both auto and manual girth welds for thickness greater than or equal to 6 mm subject to Company Approval. For thickness less than 6 mm, RT shall be carried out. The acceptance criteria for AUT is API 1104 and this specification. COMPANY will accept only experienced NDT sub-contractor experienced in the pipeline AUT for at-least 5 years. AUT procedure (including calibration and validation of the AUT) is subject to approval by COMPANY. Where CONTRACTOR uses RT, the acceptance criteria shall be as per DGS.

Evaluation shall be Auto UT based criteria specified in API 1104 for both auto weld and manual weld. Radiography is not warranted where Auto UT is carried out.

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15.4 MAGNETIC PARTICLE INSPECTION

CONTRACTOR shall supply a Magnetic Particle Inspection (MT) procedure for review and approval by COMPANY. The prod type technique per ASME Section V, Article 7, T-743 is not permitted.

Magnetic particle inspection is required for the following:

a) Re-bevelling due to pipe cutting, pipe damage and/or cut outs (not initial pipe end re-bevelling for automatic GMAW welding)
b) Repair excavations.
c) On removal of arc strikes.

MT operators shall be qualified in accordance with ASNT Level II.

COMPANY reserves the right to test proposed MT operators prior to acceptance, using the approved procedure on a test weld.

15.5 DEFECT ACCEPTANCE LEVELS

All defect acceptance limits specified below are in addition to API Standard 1104 (latest addition) Section 9; Acceptance Standard for NDT:

- The internal root penetration shall not exceed 3mm
- The external weld reinforcement shall be uniform and in no area shall the weld cap be lower than the adjacent pipe wall.
- An isolated pore greater than 25 percent of the pipe thickness or 3mm whichever is the smaller in any direction, shall be considered unacceptable.
- Cracks found by non-destructive testing in any form or orientation (excluding visual surface crater cracks) shall be unacceptable and shall result in the whole weld being cut out, edges re-bevelled and re-welded.
- The maximum permitted individual length of inadequate penetration without high-low shall be 15mm. The maximum aggregate length of defect shall be 20mm within each completed girth weld.

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- The maximum permitted individual length of inadequate penetration due to high-low shall be 15mm. The maximum aggregate length shall be 20mm within each completed girth weld.
- Incomplete fusion is not permitted.
- Incomplete fusion due to cold lap up to max of 15 mm is allowed.
- Burn-through is not acceptable. All burn-through shall be repaired to the satisfaction of COMPANY.
- The aggregate length of all indications of Hollow-Bead Porosity (HB) shall not exceed 3% of the weld length.
- Tungsten inclusions shall be treated as individual inclusions and shall not exceed 2 per weld and 3mm in size separated by a minimum distance of 50mm.
- No arc strikes are permitted outside the weld bead. Repair by welding of arc strikes is not permitted.
- Removal of arc strikes shall be at the discretion of COMPANY. When removal is allowed it shall be done by careful grinding only. The ground area shall be examined by magnetic particle inspection to ensure freedom from defects and the remaining wall thickness shall be checked by ultrasonic testing. If grinding to remove the effects of arc strikes reduces the wall thickness below the specified minimum wall thickness of the pipe, the defective pipe section shall be cut off.
- Where Magnetic Particle Inspection is applied on parent pipe bevels following re-bevelling, no linear defects or laminations are permitted. Any indications revealed by MT shall be investigated and removed by grinding or re-cutting, and again re-bevelling.
- Any metal inclusions of materials other than tungsten shall be repaired.

Each NDT method shall have a documented procedure submitted for Company's review and acceptance.

16.0 REPAIR OF WELDS

The system for reporting defects shall be such that defects can be identified quickly and are accurately located.

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All repairs shall be carried out in accordance with API 1104 (latest addition) section 10; Repair & Removal of Defects and shall be approved by the company.

The decision to proceed with repair in each instance shall be approved by COMPANY. Only qualified, COMPANY approved repair procedures shall be used.

Defects shall be removed by grinding, machining or arc-air gouging only. Arc-air gouging shall be performed in accordance with procedures approved by COMPANY and only by personnel who are competent to perform such work.

The cavity shall be finished by grinding to a bright and smooth finish with a tapered entry and exit profile. The length of the cavity shall be 50 mm minimum.

The cavity shall be subjected to magnetic particle inspection after grinding.

All repairs shall be subject to the same level of NDE as the original weld. The NDE shall extend to plus 75mm beyond each end of the repair.

Only one repair attempt shall be made. Should the repair weld fail to meet the acceptance requirements the whole weld shall be cut-out, edges re-beveled and re-welded.

Weld repairs to the pipe body or to the body of fittings or bevels are not permitted.

Single pass repairs are not permitted.

Repair to the weld root by welding from the inside of the pipe is not permitted. Repair procedures shall be qualified in accordance with paragraph 12.4.

As per COMPANY KPI, weld repairs should be less than 5% on joint basis, in the event when weld repairs approach the 5% mark the contractor shall submit a justification stating the reasons for the weld repairs and remedial action needed to prevent any excessive repairs for company review and comments.

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17.0 OTHER WELDING

No cutting or welding, other than that required to perform the permanent work, will be allowed without written permission from COMPANY.

18.0 WELDING MATERIAL DISPOSAL

CONTRACTOR shall furnish each welder with a container for disposal of weld rod stubs. Containers shall be emptied at locations acceptable to COMPANY. CONTRACTOR shall clean welding areas at least once per day of all trash, waste, rod stubs, etc. to the satisfaction of the COMPANY.

19.0 RECORDS

Permanent records shall be made, with full supporting information of all welds, including repairs and re-welds. Records shall include a pipe tracking system or field drawing which identifies the final location co-ordinates of each pipe and weld.

All welding repair rates shall be reported to COMPANY weekly per spread and per welder/auto machine with an analysis of the type of defects occurring. This shall be presented in a software package format e.g., excel/access.

In addition to hard copies, all project welding procedures (WPS and PQR) shall be furnished on CD ROM format with a suitable COMPANY approved Welding software package., e.g. Welding Institute Weld-Spec Program or other equivalent industry accepted standard package.

20.0 ATTACHMENTS

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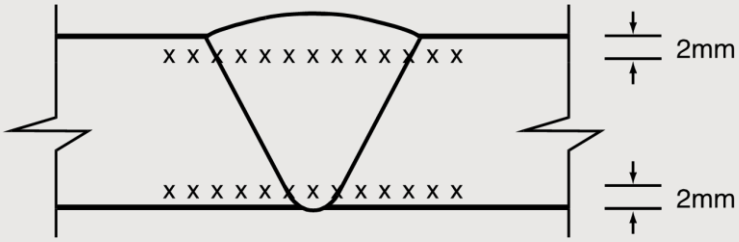
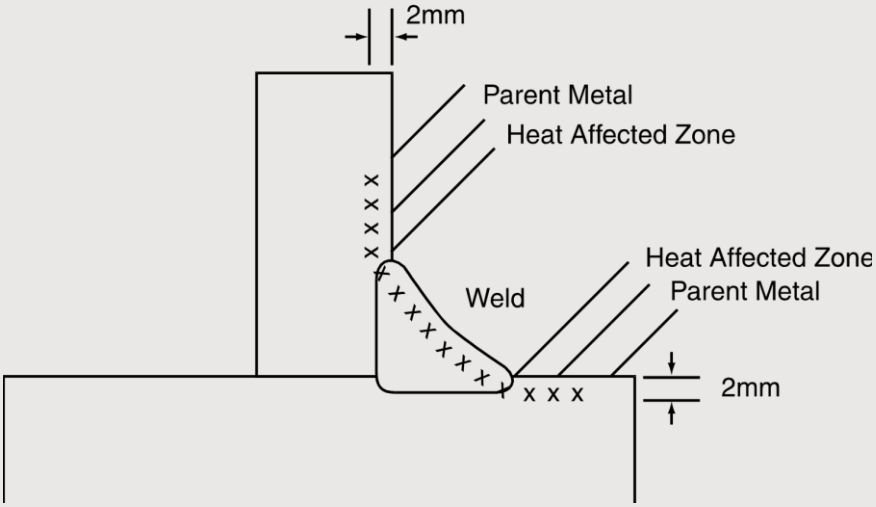
ATTACHMENT 1: VICKERS HARDNESS MEASUREMENT TRAVERSES

The Vickers hardness testing shall be carried out as follows:

- The load shall be 10Kg.
- Indentations shall be located along the axes as shown below.
- At least 6 equally spaced indentations are to be made in the weld metal.
- The heat affected zone shall be checked from base metal to weld metal and the distance between indentations shall be 0.5mm.
- One heat affected zone indentation shall be within 0.5mm of the fusion line on each side of the weld.





*Indentations shall also be made on the root side of the weld (weld metal and heat affected zone).

FIGURE 1

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ATTACHMENT 2: IMPACT TESTING REQUIREMENTS

Specimens and notch detail per ASTM A 370

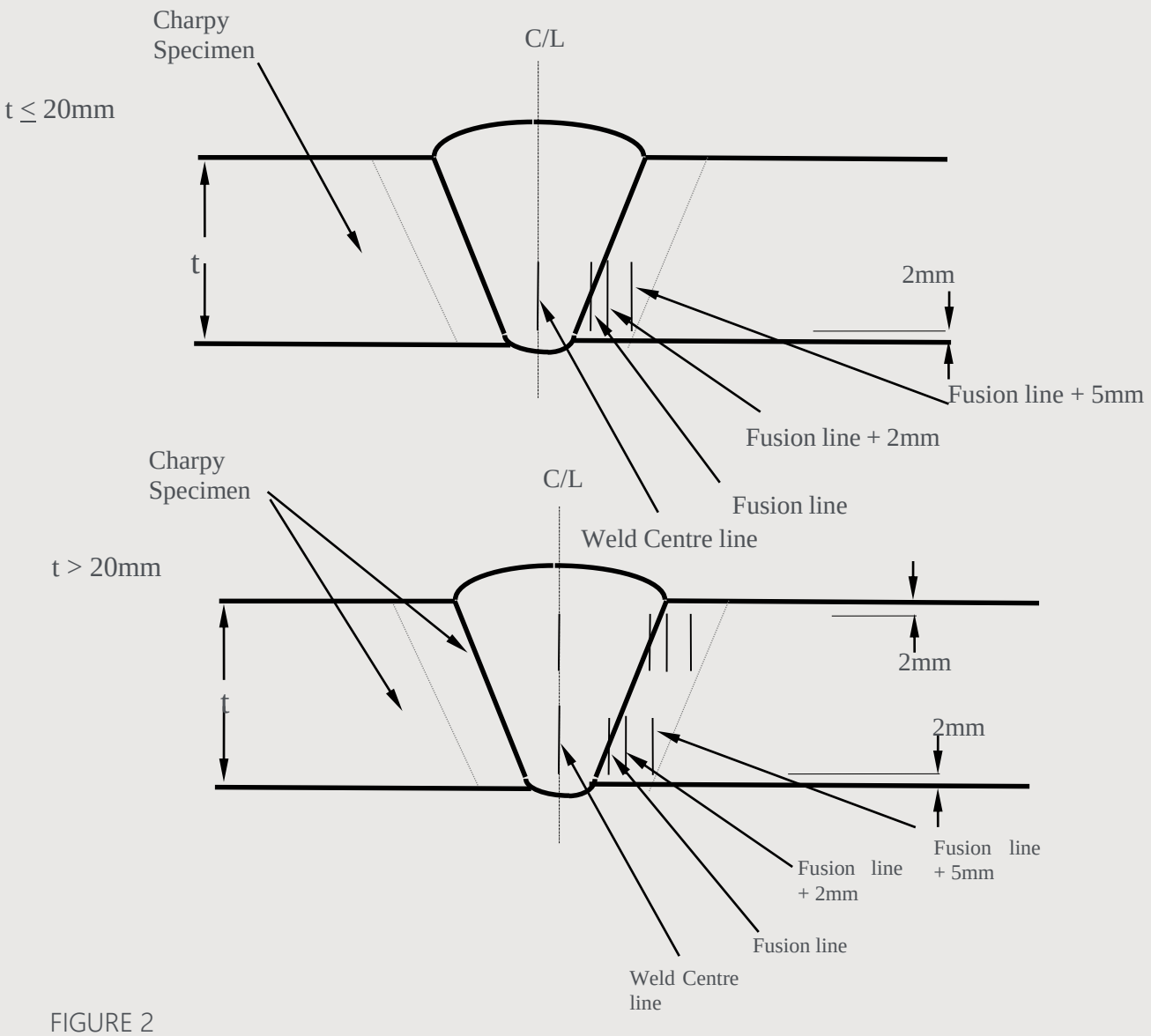

FIGURE 2

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ATTACHMENT 3: LOCATION OF PARTIAL PENETRATION WELD REPAIRS

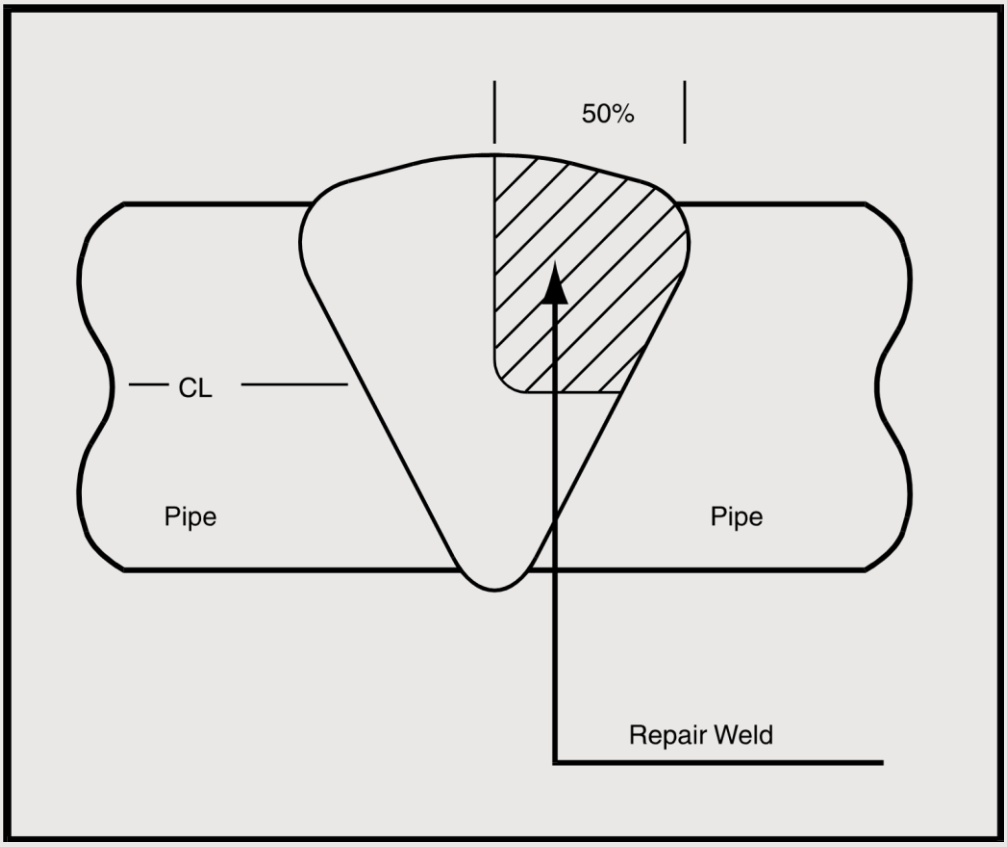

FIGURE 3

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